

8.03: THE PHYSICS OF WAVES

MARC D NICHİTU | MASSACHUSETTS INSTITUTE OF TECHNOLOGY | FALL 2025

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This document is a summary of the course 8.03 Physics III: Waves, taught in Fall 2025 by Professor Riccardo Comin of the Massachusetts Institute of Technology. This document was created for the sole purpose of exam review for this course; it uses notes written by Professors Alex Shvonski, Vladan Vuletic, Or Hen, and others who contributed to the online MITx notes.

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External Sources

- Georgi, Howard. *The Physics of Waves*. Prentice Hall, 1993
- Purcell, Edward. *Electricity and Magnetism*. Cambridge University Press, 2013

1 Simple Systems: One Mass

1.1 The Harmonic Oscillator

We begin with Newton's laws. In particular, for a mass m ,

$$\mathbf{F}_{\text{net}} = m\mathbf{a}_{\text{net}} \quad (1.1)$$

Let us write the acceleration as the second derivative of position:

$$\mathbf{F}_{\text{net}} = m\ddot{\mathbf{x}}_{\text{net}} \quad (1.2)$$

If we work in only one dimension, $\mathbf{e}_1$ for example, our equation simplifies:

$$F_{\text{net}} = m\ddot{x} \quad (1.3)$$

Consider a block of mass m attached to a fixed wall with a spring having constant k . Suppose there is no friction. Consider a coordinate system such that the origin is located at the equilibrium position of the spring. Thus, Newton's Second Law becomes

$$-kx = m\ddot{x} \Rightarrow \ddot{x} + \frac{k}{m}x = 0 \quad (1.4)$$

We generalize this type of equation as a *simple harmonic oscillator*, defined below.

Definition 1.1.1 (SHO)

The *Simple Harmonic Oscillator* (SHO) is a system of a mass m described by a differential equation having the following form,

$$\ddot{x} + \omega_0^2 x = 0 \quad (1.5)$$

where x describes the difference between the position of the mass as a function of time and its equilibrium position.

We can justify the choice of the variable ω_0^2 as the coefficient of x by solving this differential equation.

Proposition 1.1.2 (Solution of the SHO)

Let a system be described by

$$\ddot{x} + \omega_0^2 x = 0 \quad (1.6)$$

The solution of the system (thus the position of the mass as a function of time $x(t)$) is given by

$$x(t) = A \cos(\omega_0 t + \varphi) \quad (1.7)$$

where A is an amplitude, found from initial conditions, and φ is a phase, also found from initial conditions.

Proof

We try the eigenfunction of the derivative operator, namely

$$x(t) = A_0 \exp(rt) \quad (1.8)$$

to observe that

$$A_0 r^2 \exp(rt) + \omega_0^2 A_0 \exp(rt) = 0 \Rightarrow r^2 = -\omega_0^2 \quad (1.9)$$

From this we observe that the two choices for r are $\pm i\omega_0$, from which we see that

$$\tilde{x}(t) = A_1 \exp(i\omega_0 t) + A_2 \exp(-i\omega_0 t) \quad (1.10)$$

A_1, A_2 are complex; we seek a real solution, and thus from the 4 initial degrees of freedom we eliminate two (complex) to keep two, which leads to a sum of cosine and sine:

$$x(t) = A_3 \sin(\omega_0 t) + A_4 \cos(\omega_0 t) \quad (1.11)$$

Let us define $A \equiv \sqrt{A_3^2 + A_4^2}$. Let us also define $\varphi = \arg(A_4 + iA_3)$. We can thus condense the two trigonometric functions into a single one:

$$x(t) = A \cos(\omega_0 t + \varphi) \quad (1.12)$$

exactly the desired result. $\square$

We thus see what the role of the coefficient of x is: it is the square of the (natural) frequency of oscillation of the system.

1.1.1 Energy

We define the *kinetic* energy of a mass as follows:

$$K \equiv \frac{m\dot{x}^2}{2} \quad (1.13)$$

The potential energy is defined as

$$U' \equiv -F \Rightarrow U = - \int F dx = \frac{kx^2}{2} \quad (1.14)$$

for a system with a simple spring. We immediately see that energy is conserved:

$$E_{\text{tot}} = K + U = \frac{m\dot{x}^2}{2} + \frac{kx^2}{2} = \frac{1}{2} (mA^2\omega_0^2 \sin^2(\omega_0 t + \varphi) + kA^2 \cos^2(\omega_0 t + \varphi)) = \frac{A^2 k}{2} \quad (1.15)$$

because $m\omega_0^2 = \cancel{m} \cdot k / \cancel{m} = k$ from equation (1.4). Because this total energy E_{tot} does not vary in time, it is conserved by the system, and thus there is no energy loss.

1.1.2 Approximations of a System

Let a system be such that the potential associated with a generalized coordinate q is given by $V(q)$. Let q_0 be a position of the system such that

$$V'(q_0) = 0 \quad \text{and} \quad V''(q_0) > 0 \quad (1.16)$$

This requires that q_0 be an *equilibrium point* of the system, and that this point represents a *stable* equilibrium. Thus, the force associated with this potential V is

$$\mathbf{F} = -\nabla V \Rightarrow F = -V' \quad (1.17)$$

Let $\delta \equiv q - q_0$. If we use a Taylor Series around q_0 , we obtain

$$V(q) = V(q_0) + \cancel{V'(q_0)\delta} + \frac{V''(q_0)}{2}\delta^2 + \mathcal{O}(\delta^3) \quad (1.18)$$

$$\Rightarrow F(q) = -V'(q) \quad (1.19)$$

$$= -\cancel{V'(q_0)} - V''(q_0)\delta + \mathcal{O}(\delta^2) \quad (1.20)$$

$$\simeq -V''(q_0)\delta \quad (1.21)$$

We thus observe that $V''(q_0)$ is an k_{eff} , if we model this arbitrary system as a mass and a spring. Thus, we can generalize the frequency as well, by observing that $\ddot{q} = \ddot{\delta}$:

$$0 = m\ddot{q} + V''(q_0)\delta \quad (1.22)$$

$$= m\ddot{\delta} + V''(q_0)\delta \quad (1.23)$$

$$\Rightarrow 0 = \ddot{\delta} + \frac{V''(q_0)}{m}\delta \quad (1.24)$$

$$\Rightarrow \delta(t) = A \exp \left(i \left(\sqrt{\frac{V''(q_0)}{m}} \right) t \right) \quad (1.25)$$

Thus, the frequency is indeed

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{V''(q_0)}{m}} \quad (1.26)$$

for $k_{\text{eff}} = V''(q_0)$.

When is this approximation no longer valid? We must return to the term

$$\mathcal{O}(\delta^3) \quad (1.27)$$

Thus, it would be necessary that the second-order approximation be sufficient, equivalent to the following expression:

$$|V''(q_0)| \gg \left| \frac{V'''(q_0)}{2} \delta \right| \Rightarrow \delta \ll 2 \left| \frac{V''(q_0)}{V'''(q_0)} \right| \quad (1.28)$$

Proposition 1.1.3 (Modeling an Arbitrary System as an SHO)

Let a system with mass m have a potential $V(q)$ as a function of a generalized coordinate $q(t)$. It is possible to model this system as an SHO around a stable equilibrium q_0 such that $V'(q_0) = 0$ and $V''(q_0) > 0$ for oscillations having the following form with the following bound on the amplitude:

$$q(t) \simeq q_0(t) + A \exp \left(i \left(\sqrt{\frac{V''(q_0)}{m}} \right) t \right) \quad ; \quad A \ll 2 \left| \frac{V''(q_0)}{V'''(q_0)} \right| \quad (1.29)$$

We are now ready to add a term to the differential equation, in the form of damping.

1.2 The Damped Harmonic Oscillator

We now consider an oscillator with a dependence on velocity, $\dot{x}$:

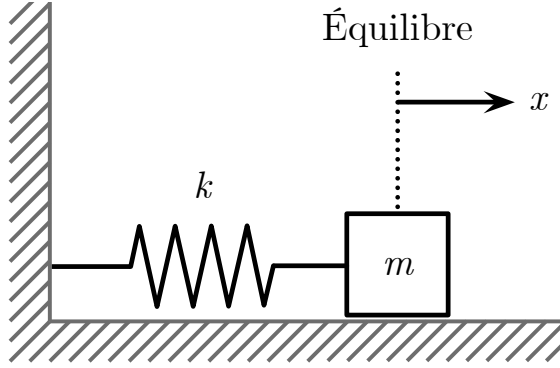


Figure 1. Undamped Oscillator

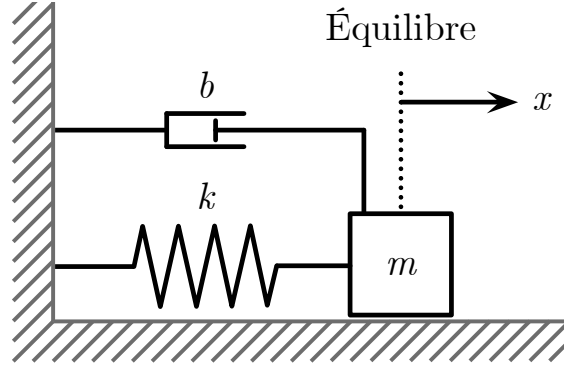


Figure 2. Damped Oscillator

The additional piece in the image on the right represents a damper, such that its force on the mass m is

$$F_{\text{damping}} = -b\dot{x} \quad (1.30)$$

Thus, the sum of forces becomes

$$m\ddot{x} = -b\dot{x} - kx \Rightarrow \ddot{x} + \frac{b}{m}\dot{x} + kx = 0 \quad (1.31)$$

We write the general form of such a system below.

Definition 1.2.1 (DHO)

The *Damped Harmonic Oscillator* is a system of a mass m described by a differential equation having the following form,

$$\ddot{x} + \Gamma\dot{x} + \omega_0^2 x = 0 \quad (1.32)$$

where x describes the difference between the position of the mass as a function of time and its equilibrium position.

The characteristic polynomial P of this differential equation allows us to see the types of permitted solutions:

$$P(D)x(t) = 0 : P(r) = 0 \Rightarrow x(t) = c_1 \exp(rt) \text{ is a solution} \quad (1.33)$$

In our case, P takes the value

$$P(r) = r^2 + \Gamma r + \omega_0^2 \quad (1.34)$$

The roots are given by

$$r = \frac{-\Gamma \pm \sqrt{\Gamma^2 - 4\omega_0^2}}{2} = -\frac{\Gamma}{2} \pm \sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \quad (1.35)$$

We thus see that there exist three main cases:

- An *overdamped* case in which the roots r_1, r_2 are distinct and real: this requires

$$\Gamma^2 > 4\omega_0^2 \quad (1.36)$$

It follows that this case would be found in situations of high damping (quantified by Γ^2).

- An *underdamped* case in which the roots r_1, r_2 are complex conjugates: this requires

$$\Gamma^2 < 4\omega_0^2 \quad (1.37)$$

It follows that this case would be found in situations of low damping, with the limit $\Gamma \rightarrow 0$ described by the SHO: $r = \pm i\omega_0$.

- A *critically damped* case in which the roots r_1, r_2 are equal (and thus real): this requires

$$\Gamma^2 = 4\omega_0^2 \quad (1.38)$$

We explore the solutions $x(t)$ associated with each case.

1.2.1 Overdamped

A system is said to be *overdamped* if

$$\Gamma^2 > 4\omega_0^2 \quad (1.39)$$

Thus,

$$r_1 = -\frac{\Gamma}{2} + \sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \quad ; \quad r_2 = -\frac{\Gamma}{2} - \sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \quad (1.40)$$

and the solution $x(t)$ takes the form

$$x(t) = c_1 \exp(r_1 t) + c_2 \exp(r_2 t) \quad (1.41)$$

The constants c_1, c_2 are found from initial conditions (typically given by the values $x(0), \dot{x}(0)$: the initial position and velocity).

An example of the response $x(t)$ is found below:

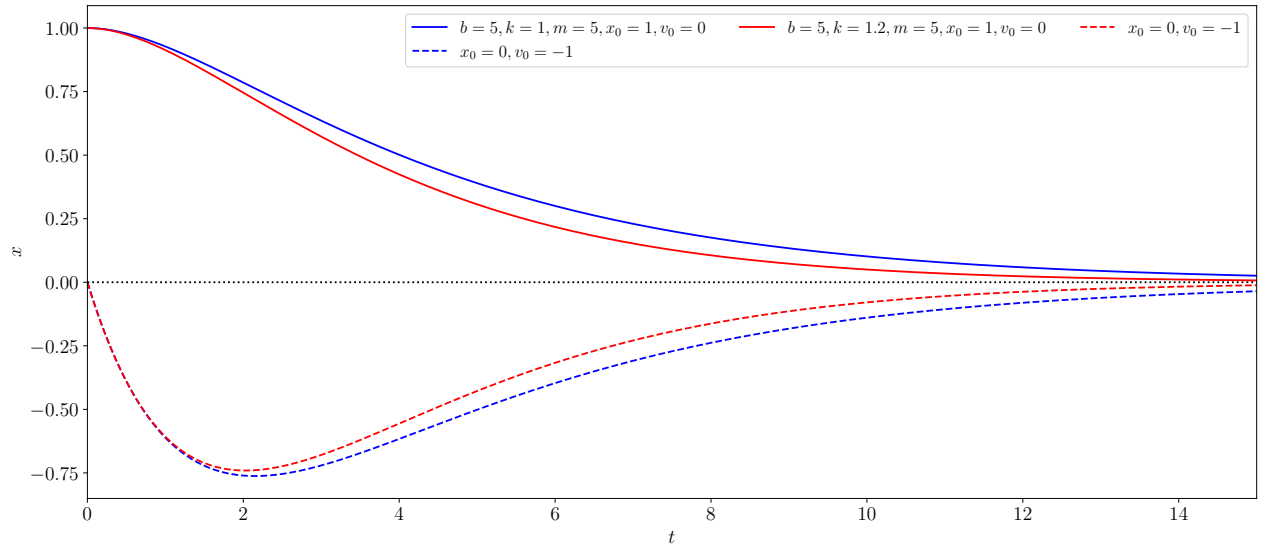


Figure 3. Solutions $x(t)$ for $\ddot{x} + \dot{x} + \frac{x}{5} = 0$ (in blue) and $\ddot{x} + \dot{x} + \frac{1.2x}{5} = 0$ (in red). The solid and dashed curves represent different initial cases: solid begins with $x(0) \neq 0; \dot{x}(0) = 0$, and the dashed begins with the opposite, thus $x(0) = 0; \dot{x}(0) \neq 0$.

If $r_1, r_2 < 0$, the system will be stable (and thus the limit as time $t \rightarrow \infty$ will be 0).

1.2.2 Underdamped

A system is said to be *underdamped* if

$$\Gamma^2 < 4\omega_0^2 \quad (1.42)$$

Thus,

$$r_1 = -\frac{\Gamma}{2} + i\sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \quad ; \quad r_2 = -\frac{\Gamma}{2} - i\sqrt{\frac{\Gamma^2}{4} - \omega_0^2} \quad (1.43)$$

It is convenient to define $\sigma \equiv -\Gamma/2$ and $\omega = \sqrt{\frac{\Gamma^2}{4} + \omega_0^2}$ such that

$$r_1 = \sigma + i\omega \quad ; \quad r_2 = \sigma - i\omega \quad (1.44)$$

Thus, for A, φ found from initial conditions, the real part of the solution $x(t)$ is

$$x(t) = Ae^{\sigma t} \cos(\omega t + \varphi) \quad (1.45)$$

This solution is stable if $\sigma < 0 \iff \Gamma > 0$. An example is drawn below:

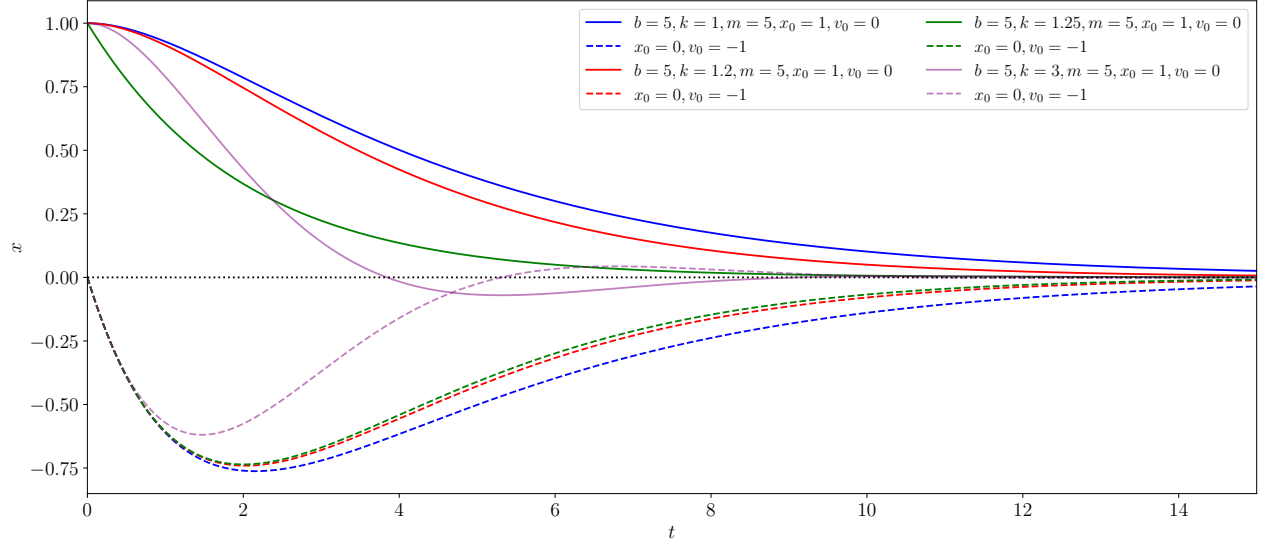


Figure 4. Solutions $x(t)$ for $\ddot{x} + \dot{x} + \frac{3x}{5} = 0$ (in pink). The solid and dashed curves represent different initial cases: solid begins with $x(0) \neq 0; \dot{x}(0) = 0$, and the dashed begins with the opposite, thus $x(0) = 0; \dot{x}(0) \neq 0$. In green are drawn curves representing critical damping.

It is also possible to rewrite the DHO equation using a *damping coefficient*, ζ :

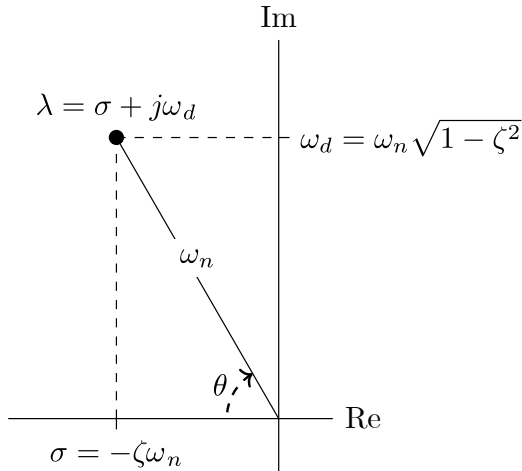


Figure 5. Poles of an underdamped system.

Definition 1.2.2 (Damping Ratio)

Let a system be an underdamped DHO having a characteristic polynomial P with complex roots $r_{1,2} = \sigma \pm j\omega$. The governing differential equation thus takes the form

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x \quad (1.46)$$

The *natural frequency* of the system is found from the damped frequency:

$$\omega_0 = \sqrt{\sigma^2 + \omega^2} = |r_1| = |r_2| \quad (1.47)$$

The *damping ratio* is

$$\zeta \equiv \frac{\sigma}{\omega_0} \quad (1.48)$$

One final characterization of such systems exists, namely the *quality factor* Q :

$$Q \equiv \frac{1}{2\zeta} \quad (1.49)$$

for ζ defined as above.

1.2.3 Critical Damping

Critical damping occurs when the roots are equal:

$$r_1 = r_2 \Rightarrow x(t) = (c_1 + c_2 t)e^{rt} \quad (1.50)$$

This descends most rapidly toward equilibrium - it is thus the desirable state for a physical system that serves to dampen shocks. We thus summarize the cases: in terms of ζ ,

Overdamped	Critical Damping	Underdamped	Undamped
$\zeta > 1$	$\zeta = 1$	$0 < \zeta < 1$	$\zeta = 0$
$Q < \frac{1}{2}$	$Q = \frac{1}{2}$	$Q > \frac{1}{2}$	$Q \rightarrow \infty$

1.3 The Driven DHO

Often, physical systems are *driven* - this indicates an external force acting on the system. If the force is constant, the equilibrium position is merely displaced (for example, consider the vertical spring, in which the constant external force would be gravity mg). However, if the force varies in time, we are forced to consider the general case:

$$\ddot{x} + \Gamma \dot{x} + \omega_0^2 x = \frac{F(t)}{m} \quad (1.51)$$

We apply the linearity of the differential operator to demonstrate that the general solution $x(t)$ for such a driven system would be the sum of the so-called “homogeneous” solution (that of the undriven system) and the so-called “particular” solution (that of the driven system without transient effects):

$$x = x_h + x_p : \begin{cases} \ddot{x}_h + \Gamma \dot{x}_h + \omega_0^2 x_h = 0 \\ \ddot{x}_p + \Gamma \dot{x}_p + \omega_0^2 x_p = F(t)/m \end{cases} \quad (1.52)$$

Consider the case of a force $F(t)$ that varies sinusoidally, with an angular frequency ω_f :

$$F(t) = A_F \cos(\omega_f t) \quad (1.53)$$

Thus, we “complexify” the force:

$$\tilde{F}(t) = A_F e^{i\omega_f t} \quad (1.54)$$

This allows us to try the general solution $x_p = A_p \exp(i\omega_f t - i\varphi)$:

$$\frac{F(t)}{m} = \ddot{x}_p + \Gamma \dot{x}_p + \omega_0^2 x_p \quad (1.55)$$

$$\Rightarrow \frac{A_F}{mA_p} \cancel{A_p e^{i\omega_f t - i\varphi}} e^{i\varphi} = -\omega_f^2 \cancel{A_p e^{i\omega_f t - i\varphi}} + i\omega_f \Gamma \cancel{A_p e^{i\omega_f t - i\varphi}} + \omega_0^2 \cancel{A_p e^{i\omega_f t - i\varphi}} \quad (1.56)$$

$$\Rightarrow \frac{A_F}{mA_p} e^{i\varphi} = \omega_0^2 - \omega_f^2 + i\omega_f \Gamma \quad (1.57)$$

We see that this represents a decomposition of the complex number $\omega_0^2 - \omega_f^2 + i\omega_f \Gamma$ into a magnitude $A_F/(mA_p)$ and a phase $-\varphi$:

$$A_p = \frac{A_F/m}{\sqrt{(\omega_0^2 - \omega_f^2)^2 + \Gamma^2 \omega_f^2}} \quad (1.58)$$

$$\varphi = \arg(\omega_0^2 - \omega_f^2 + i\omega_f \Gamma) \quad (1.59)$$

We can see what driving frequency ω_f leads to the largest amplitude A_p by finding the minimum of the denominator:

$$0 = \frac{\partial}{\partial \omega_f} ((\omega_0^2 - \omega_f^2)^2 + \Gamma^2 \omega_f^2) \quad (1.60)$$

$$= 2(\omega_0^2 - \omega_f^2)(-2\omega_f) + 2\omega_f \Gamma^2 \quad (1.61)$$

$$= 2\omega_f \Gamma^2 - 4\omega_f (\omega_0^2 - \omega_f^2) \quad (1.62)$$

$$\Rightarrow \Gamma^2 = 2(\omega_0^2 - \omega_f^2) \quad (1.63)$$

$$\Rightarrow \omega_f = \sqrt{\omega_0^2 - \frac{\Gamma^2}{2}} \quad (1.64)$$

This amplitude is that of *resonance*. If $\Gamma = 0$, the natural frequency would make the denominator zero, which would lead to an unbounded amplitude, which is not real (in life there is always some damping). The phase will sweep between 0 and π , being $\pi/2$ at resonance.

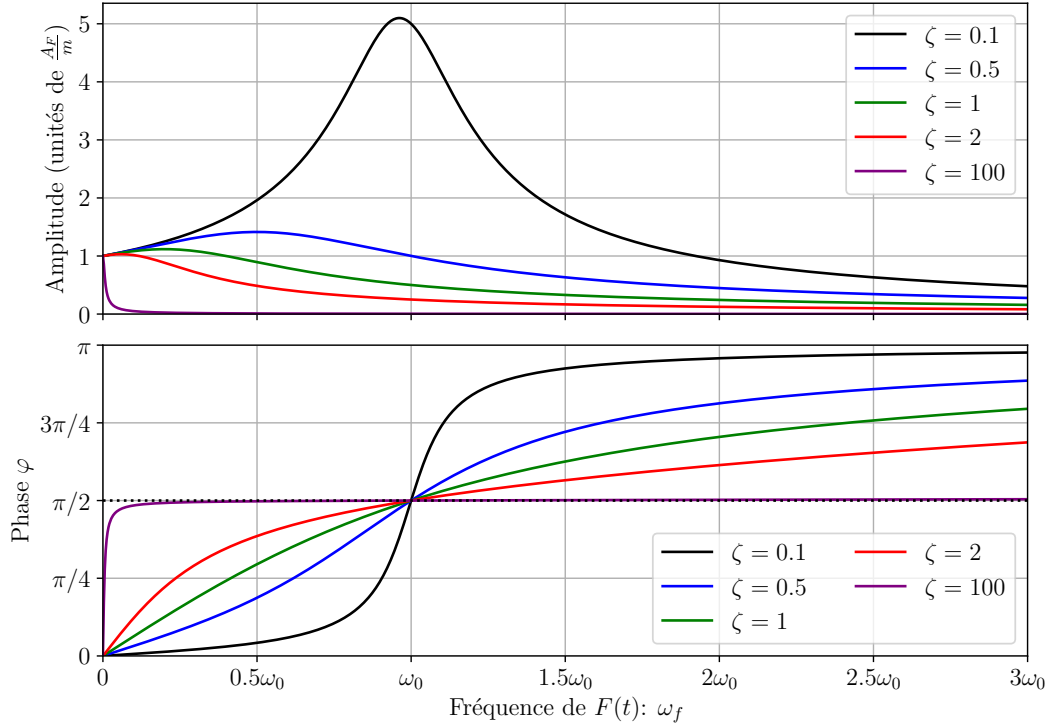


Figure 6. Amplitude and Phase under forces $F(t)$ having a frequency $\omega_f \in [0, 3\omega_0]$.

We can thus characterize a system given its *frequency response*, represented above. If the growth of the slope of the phase, $\varphi''(\omega_f)$, is positive at low frequency ($\omega_f \ll \omega_0$), the system is *underdamped* and $\zeta \in (0, 1)$. If the second derivative $\varphi''(\omega_f)$ is zero in this regime, $\zeta = 1$ and thus the system is characterized by *critical damping*. If the second derivative is negative, $\zeta > 1$ and thus the system is *overdamped*. The effect of resonance is found in the amplitude graph – clearly, the height of the peak is greater and more pronounced for $\zeta \sim 0$ than for $\zeta \sim 1$. For $\zeta = 1$, there is a small peak that remains; however for $\zeta \gg 1$, there is no longer a peak and the system behaves like a very aggressive low-pass filter.

We are now ready to analyze systems of such oscillators.

2 Systems of Oscillators

2.1 Normal Modes

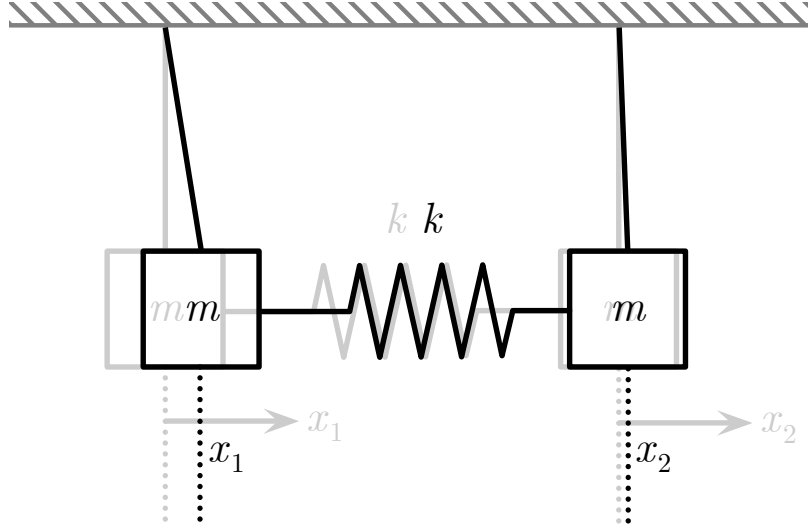


Figure 7. System of two masses as pendulums, connected with a spring.

Consider the system drawn above. Let us write the differential equations that describe the behavior of each mass. We simplify the modeling of the system by ignoring vertical displacements. Using $\sin \theta \simeq \theta$, the gravitational torque takes the value

$$\tau = I\ddot{\theta} \Rightarrow mgl \sin \theta = -ml^2\ddot{\theta} \Rightarrow \ddot{\theta} = -\frac{g}{l} \sin \theta \simeq -\frac{\theta g}{l} \quad (2.1)$$

Thus, because $x = l \sin \theta \simeq l\theta$, we have that the equation of motion for a simple pendulum is

$$\ddot{x} + \frac{g}{l}x = 0 \quad (2.2)$$

Now we can write the equations for the two masses. Let x_1 be the displacement between the equilibrium position of the mass on the left and its current position; let x_2 be defined equivalently for the mass on the right. Thus,

$$m\ddot{x}_1 = -\frac{mg}{l}x_1 - kx_1 + kx_2 \quad (2.3)$$

$$m\ddot{x}_2 = -\frac{mg}{l}x_2 - kx_2 + kx_1 \quad (2.4)$$

We write these equations in vector form, observing that the dependencies on x_1 and x_2 are linear:

$$\ddot{\mathbf{x}} = \frac{d^2}{dt^2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\frac{g}{l} - \frac{k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{g}{l} - \frac{k}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -\mathbf{M}^{-1}\mathbf{K}\mathbf{x} \quad (2.5)$$

where

$$\mathbf{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad ; \quad \mathbf{K} = \begin{bmatrix} \frac{mg}{l} + k & -k \\ -k & \frac{mg}{l} + k \end{bmatrix} \quad (2.6)$$

We seek oscillatory solutions; thus, we propose a solution having the form

$$\mathbf{x} = \mathbf{A}e^{i\omega t} \quad (2.7)$$

If we try to find constraints on $\mathbf{A}$ and ω by taking the second derivative, we obtain that

$$\ddot{\mathbf{x}} = -\omega^2 \mathbf{A}e^{i\omega t} = -\omega^2 \mathbf{x} \quad (2.8)$$

and thus

$$\omega^2 \mathbf{x} = \mathbf{M}^{-1} \mathbf{K} \mathbf{x} \quad (2.9)$$

Thus, ω^2 is an eigenvalue of $\mathbf{M}^{-1} \mathbf{K}$, and $\mathbf{A}$ will be the associated eigenvector. Thus, we seek these ω s:

$$0 = |\mathbf{M}^{-1} \mathbf{K} - \omega^2 \mathbf{I}| = \begin{vmatrix} \frac{g}{l} + \frac{k}{m} - \omega^2 & -\frac{k}{m} \\ -\frac{k}{m} & \frac{g}{l} + \frac{k}{m} - \omega^2 \end{vmatrix} = \left(\frac{g}{l} + \frac{k}{m} - \omega^2 \right)^2 - \left(-\frac{k}{m} \right)^2 \quad (2.10)$$

We know from algebra that

$$a^2 - b^2 = 0 \iff a = \pm b \quad (2.11)$$

Thus,

$$\frac{g}{l} + \frac{k}{m} - \omega^2 = \pm \frac{k}{m} \implies \omega^2 = \frac{g}{l} + \frac{k}{m} \pm \frac{k}{m} = \left\{ \frac{g}{l}, \frac{g}{l} + \frac{2k}{m} \right\} \quad (2.12)$$

Thus, the permitted frequencies are

$$\omega_1 = \sqrt{\frac{g}{l}} \quad \text{or} \quad \omega_2 = \sqrt{\frac{g}{l} + \frac{2k}{m}} \quad (2.13)$$

We can find the associated eigenvectors:

$$\mathbf{M}^{-1} \mathbf{K} \mathbf{A}_1 = \omega_1^2 \mathbf{A}_1 \iff \mathbf{0} = (\mathbf{M}^{-1} \mathbf{K} - \omega_1^2 \mathbf{I}) \mathbf{A}_1 = \begin{bmatrix} \frac{k}{m} & -\frac{k}{m} \\ -\frac{k}{m} & \frac{k}{m} \end{bmatrix} \mathbf{A}_1 \Rightarrow \mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (2.14)$$

$$\mathbf{M}^{-1} \mathbf{K} \mathbf{A}_2 = \omega_2^2 \mathbf{A}_2 \iff \mathbf{0} = (\mathbf{M}^{-1} \mathbf{K} - \omega_2^2 \mathbf{I}) \mathbf{A}_2 = \begin{bmatrix} -\frac{k}{m} & -\frac{k}{m} \\ -\frac{k}{m} & -\frac{k}{m} \end{bmatrix} \mathbf{A}_2 \Rightarrow \mathbf{A}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (2.15)$$

Thus, the solution $\mathbf{x}(t)$ would be

$$\mathbf{x}(t) = \text{Re} \left(\sum c_i \mathbf{A}_i \exp(i\omega_i t) \right) = \text{Re} \left(c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \exp \left(i \sqrt{\frac{g}{l}} t \right) + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \exp \left(i \sqrt{\frac{g}{l} + \frac{2k}{m}} t \right) \right) \quad (2.16)$$

$$= \begin{bmatrix} c_1 \cos \left(t \sqrt{\frac{g}{l}} \right) + c_2 \cos \left(t \sqrt{\frac{g}{l} + \frac{2k}{m}} \right) \\ c_1 \cos \left(t \sqrt{\frac{g}{l}} \right) - c_2 \cos \left(t \sqrt{\frac{g}{l} + \frac{2k}{m}} \right) \end{bmatrix} \quad (2.17)$$

We call these vectors $\mathbf{A}_i$ *normal modes*.

Definition 2.1.1 (Normal Mode)

Let a system be described by the differential equation $\ddot{\mathbf{x}} = -\mathbf{M}^{-1} \mathbf{K} \mathbf{x}$.

The eigenvectors $\mathbf{A}_i$ of the matrix $-\mathbf{M}^{-1} \mathbf{K}$ are the *normal modes* of the system.

For our two-block system in Figure 7, these modes can be interpreted as “fundamental motions” – the mode $\mathbf{A}_1 = \langle 1, 1 \rangle$ represents a motion of both masses in the same direction (in particular, to the right); the mode $\mathbf{A}_2 = \langle 1, -1 \rangle$ represents a motion of both masses in opposite directions, both toward the interior.

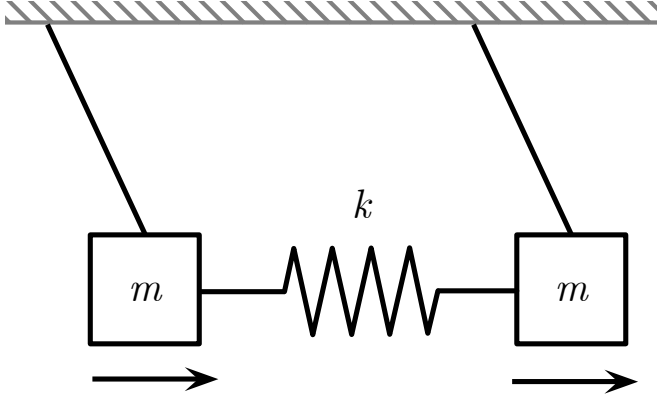


Figure 8. Normal mode $\mathbf{A}_1 = \langle 1, 1 \rangle$

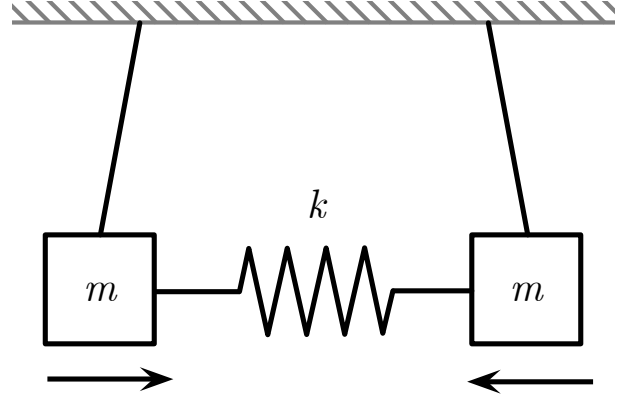


Figure 9. Normal mode $\mathbf{A}_2 = \langle 1, -1 \rangle$

It is possible to guess the normal modes by isolating portions of the system such that a spring or a group of parallel springs act. For example, consider the following system:

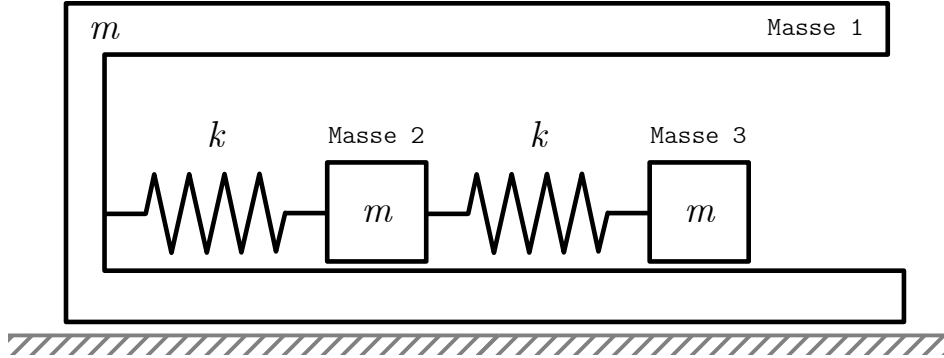


Figure 10. Nontrivial block system.

The normal modes would be

$$\mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} ; \quad \mathbf{A}_2 = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} ; \quad \mathbf{A}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad (2.18)$$

In each mode, we choose a spring or a mass that does not oscillate. The first mode is the trivial one: no spring oscillates. The second mode allows the oscillation of the left spring while keeping the right spring without deformations. The third mode keeps the second mass stationary while oscillating masses 1 and 3 around it.

2.2 Symmetries

2.2.1 Rotational

Sometimes, it is possible to find the normal modes from symmetry. Consider a system such that it is possible to replace mass m_i with mass m_{i-1} , as for example a system with rotational symmetry:

$$\begin{cases} m\ddot{\theta}_1 = -k\theta_1 + k\theta_2 + k\theta_3 \\ m\ddot{\theta}_2 = -k\theta_2 + k\theta_1 + k\theta_3 \\ m\ddot{\theta}_3 = -k\theta_3 + k\theta_2 + k\theta_1 \end{cases} \implies m \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \\ \ddot{\theta}_3 \end{bmatrix} = -k \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} \quad (2.19)$$

It is clear that

$$\mathbf{S} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \mathbf{S} = \mathbf{S} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix} \quad (2.20)$$

Thus,

$$\ddot{\mathbf{x}} = -\mathbf{M}^{-1}\mathbf{K}\mathbf{x} \iff \mathbf{S}\ddot{\mathbf{x}} = -\mathbf{M}^{-1}\mathbf{K}\mathbf{S}\mathbf{x} = -\mathbf{M}^{-1}\mathbf{S}\mathbf{K}\mathbf{x} = -\mathbf{S}\mathbf{M}^{-1}\mathbf{K}\mathbf{x} \quad (2.21)$$

From this we deduce that every normal mode $\mathbf{A}_i$, which is already an eigenvector of $\mathbf{M}^{-1}\mathbf{K}$, would also be an eigenvector of $\mathbf{S}$, because

$$\tilde{\mathbf{A}}_i \equiv \mathbf{S}\mathbf{A}_i \Rightarrow -\omega_i^2 \tilde{\mathbf{A}}_i = -\mathbf{M}^{-1}\mathbf{K}\tilde{\mathbf{A}}_i \quad (2.22)$$

but because $\mathbf{A}_i$ and $\tilde{\mathbf{A}}_i$ have the same associated frequency, they must be proportional – thus, $\mathbf{A}_i$ is an eigenvector of $\mathbf{S}$ as well:

$$\exists \alpha : \mathbf{S}\mathbf{A}_i = \tilde{\mathbf{A}}_i = \beta \mathbf{A}_i \quad (2.23)$$

Thus, because finding a symmetry matrix $\mathbf{S}$ is much simpler than finding the eigenvectors of $\mathbf{M}^{-1}\mathbf{K}$, we can quickly deduce that for

$$\mathbf{S} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (2.24)$$

the normal modes are

$$\mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad ; \quad \mathbf{A}_2 = \begin{bmatrix} \exp(2i\pi/3) \\ \exp(4i\pi/3) \\ 1 \end{bmatrix} \quad ; \quad \mathbf{A}_3 = \begin{bmatrix} \exp(4i\pi/3) \\ \exp(8i\pi/3) \\ 1 \end{bmatrix} \quad (2.25)$$

because in general

$$\mathbf{S}^3 = \mathbf{I} \Rightarrow \beta^3 = 1 \Rightarrow \forall k \in \{0, 1, 2\} : \mathbf{S}\mathbf{A}_k = \beta_k \mathbf{A}_k \Rightarrow [\mathbf{A}_k]_j = \beta_k^{j-1} = \exp\left(\frac{2ik\pi}{3}\right) \quad (2.26)$$

For a rotational symmetry of order n ,

$$\mathbf{S}^n = \mathbf{I} \Rightarrow \beta^n = 1 \Rightarrow \forall k \in \{0, \dots, n-1\} : \mathbf{S}\mathbf{A}_k = \beta_k \mathbf{A}_k \Rightarrow [\mathbf{A}_k]_j = \beta_k^{j-1} = \exp\left(\frac{2ik\pi}{n}\right) \quad (2.27)$$

2.2.2 Reflection

In the simple case above (Fig. 7), drawn below, we observe a reflection symmetry, which allows us to swap x_1 with $-x_2$; the normal modes come immediately:

$$\mathbf{S} = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix} \Rightarrow \mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad ; \quad \mathbf{A}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (2.28)$$

For a system of three masses having positions x_1 , x_2 , and x_3 , with none fixed, and two springs with constant k connecting the second mass with the first and the third, the matrix $\mathbf{S}$ takes the form

$$\mathbf{S} = \begin{bmatrix} 1 & & \\ & 0 & \\ & & -1 \end{bmatrix} \quad (2.29)$$

and thus the normal modes are

$$\mathbf{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad ; \quad \mathbf{A}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad ; \quad \mathbf{A}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (2.30)$$

which can also be found using the trick of fixing masses or springs.

2.2.3 Translational

It is also possible to describe a finite translation symmetry. Consider a system of masses suspended by pendulums, each connected to its neighbors with a spring. Let a be the distance between the masses. Thus, for an infinite system,

$$S = \begin{bmatrix} \ddots & & & & \\ & \ddots & & & \\ & & 0 & 1 & \\ & & & 0 & 1 \\ & & & & \ddots & \ddots \\ & & & & & \ddots & \ddots \end{bmatrix} \quad (2.31)$$

We thus think of normal modes such that

$$\forall i \in \mathbb{Z} : \exists \beta_i \in \mathbb{C} : \mathbf{A}_i = \begin{bmatrix} \vdots \\ \beta_i^j \\ \vdots \end{bmatrix} \quad (2.32)$$

2.3 Dispersion

The matrix $M^{-1}\mathbf{K}$ would take the form

$$M^{-1}\mathbf{K} = \begin{bmatrix} \ddots & & & & \\ & \ddots & & & \\ & & -C & 2B & -C & \\ & & & -C & 2B & -C \\ & & & & \ddots & \ddots & \ddots \end{bmatrix} \quad (2.33)$$

for $B = \frac{g}{2l} + \frac{k}{m}$ and $C = \frac{k}{m}$. Thus, we obtain

$$\omega^2 \begin{bmatrix} \vdots \\ \beta_i^j \\ \vdots \end{bmatrix} = \begin{bmatrix} \ddots & & & & \\ & \ddots & & & \\ & & -C & 2B & -C & \\ & & & -C & 2B & -C \\ & & & & \ddots & \ddots & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \beta_i^j \\ \vdots \end{bmatrix} \implies \omega_i^2 \beta_i^j = -C\beta_i^{j-1} + 2B\beta_i^j - C\beta_i^{j+1} \quad (2.34)$$

$$\implies \omega_i^2 = 2B - C\beta_i^{-1} - C\beta_i \quad (2.35)$$

Let k be the *wavenumber*, such that $\beta = \exp(ika)$ (and k is a variable that accounts for the infinite number of normal modes, indexed by i); we thus obtain a dependence of the angular frequency ω on k :

$$\omega^2 = 2B - C \exp(-ika) - C \exp(ika) \quad (2.36)$$

If we write the real part of this expression, we obtain the following fact.

Proposition 2.3.1 (Dispersion of a Discrete System)

Let a system be translationally invariant (with fundamental distance a) of masses. The *dispersion* of such a system is

$$\omega^2 = 2B - 2C \cos(ka) \quad (2.37)$$

for the equation of motion for a generalized coordinate $q_j(t)$ of the mass indexed by j given by

$$\ddot{q}_j = -Cq_{j-1} + 2Bq_j - Cq_{j+1} \quad (2.38)$$

We can specialize the general dispersion given by Eq. (2.37). First, let us use the B and C for the infinite pendulum system:

$$B = \frac{g}{2l} + \frac{k}{m} \quad ; \quad C = \frac{k}{m} \quad (2.39)$$

Thus,

$$\omega^2 = \frac{g}{l} + \frac{2k}{m} (1 - \cos(ka)) \quad (2.40)$$

It is possible to simplify the expression $1 - \cos 2x$:

$$1 - \cos 2x = 1 - (1 - 2 \sin^2 x) = 2 \sin^2 x \quad (2.41)$$

Thus,

$$\boxed{\omega^2 = \frac{g}{l} + \frac{4k}{m} \sin^2 \frac{ka}{2}} \quad (2.42)$$

2.4 Beaded String

We now consider a new case, namely a string (massless) with beads (each having mass m , spaced a apart from one another) as in the image below:

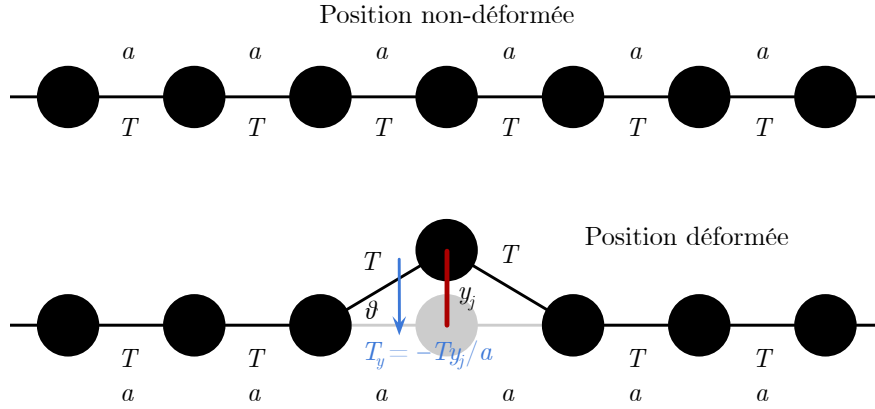


Figure 11. Beaded string.

We calculate B and C . First, assuming small deformations (not at all to scale as drawn above), we have that $\vartheta \simeq \sin \vartheta \simeq \tan \vartheta$; thus,

$$\frac{y_j}{a} = \tan \vartheta \simeq \sin \vartheta = \frac{-T_y}{T} \Rightarrow T_y = -\frac{T y_j}{a} \quad (2.43)$$

as indicated in the figure. Thus, the equation of motion for the mass indexed by j is

$$\ddot{y}_j = -\frac{T}{ma} (2y_j - y_{j+1} - y_{j-1}) \quad (2.44)$$

Thus, the dispersion for this system is

$$B = C = \frac{T}{ma} \Rightarrow \boxed{\omega^2 = \frac{4T}{ma} \sin^2 \frac{ka}{2}} \quad (2.45)$$

We can establish constraints on the permitted values of k ; we recall that $y(x) = \exp(ikx)$ and thus

$$A(x + \lambda) = A(x) \Rightarrow \exp(ikx) \exp(ik\lambda) = \exp(ikx) \Rightarrow k\lambda = 2\pi \Rightarrow \lambda = \frac{2\pi}{k} \quad (2.46)$$

Because the values of λ are limited (for a discrete system, $\exists n \in \mathbb{N} : \lambda = na$), such that the trivial solution (expressed as $y(x) = 0$) has $\lambda = a \Rightarrow k = 2\pi/a$ and the solution having the smallest wavelength (apart from the trivial one) has $\lambda = 2a \Rightarrow k = \pi/a$. Thus, we observe that the permitted values of k are

$$\forall n \in \mathbb{N} : k = \frac{2\pi}{na} ; \lim_{n \rightarrow \infty} k|_n = 0 \quad (2.47)$$

For the dispersion found above, we observe (interestingly) that $k \rightarrow 0 \Rightarrow \omega \rightarrow 0$.

If we consider a non-infinite system (having for example N masses), having boundary conditions (such as for example $y(-a/2) = 0$ and $y((N + 1/2)a) = 0$), there will be N permitted modes; they are given by the k s such that

$$y(x) = \sum_{n=1}^N A_n \cos(k|_n x) \cos(\omega|_n t) = \sum_{n=1}^N A_n \cos\left(\frac{n\pi}{Na}x\right) \cos\left(\omega\left(\frac{n\pi}{Na}\right)t\right) \quad (2.48)$$

The coefficients $\{A_n\}$ can be found from the initial conditions, for example through a Fourier Series that would describe the initial deformations using a sum of cosines. A more detailed discussion of boundary conditions and the sum of normal modes is found in the section on the continuous regime, because the same techniques apply in both regimes.

2.5 LC Circuits

A final example would be a series of chained LC circuits, as below.

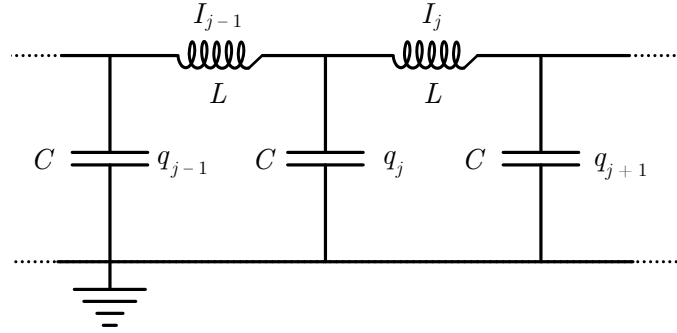


Figure 12. Chain of capacitors C and inductors L .

Let us write Kirchhoff's law for the two loops drawn:

$$\frac{q_{j-1}}{C} - L\dot{I}_{j-1} - \frac{q_j}{C} = 0 \quad ; \quad \frac{q_j}{C} - L\dot{I}_j - \frac{q_{j+1}}{C} = 0 \quad (2.49)$$

We write Kirchhoff's law for the complete junction:

$$I_{j-1} = \dot{q}_j + I_j \Rightarrow \dot{I}_{j-1} = \ddot{q}_j + \dot{I}_j \quad (2.50)$$

We combine the equations:

$$0 = q_{j-1} - LC(\ddot{q}_j + (q_j - q_{j+1})L^{-1}C^{-1}) - q_j \quad (2.51)$$

$$\Rightarrow \ddot{q}_j = -\frac{1}{LC}(2q_j + q_{j-1} + q_{j+1}) \quad (2.52)$$

Here,

$$B = C = \frac{1}{LC} \quad (2.53)$$

and thus the dispersion is

$$\boxed{\omega^2 = \frac{4}{LC} \sin^2 \frac{ka}{2}} \quad (2.54)$$

3 Limit to the Continuous Regime

3.1 The Wave Equation

Let us return to the case of the beaded string. Let us define the *linear density* μ such that

$$\mu \equiv \frac{m}{a} \Rightarrow m = \mu a \quad (3.1)$$

Let us write the governing differential equation using μ :

$$\ddot{y}_j = -\frac{T}{ma}(2y_j - y_{j+1} - y_{j-1}) = -\frac{T}{\mu a^2}(2y_j - y_{j+1} - y_{j-1}) \quad (3.2)$$

Let us take the limit as $a \rightarrow 0$:

$$\lim_{a \rightarrow 0} \frac{2y_j - y_{j+1} - y_{j-1}}{a^2} = \lim_{a \rightarrow 0} \frac{1}{a} \left(\frac{y_j - y_{j+1}}{a} - \frac{y_{j-1} - y_j}{a} \right) \quad (3.3)$$

$$= \lim_{a \rightarrow 0} \frac{1}{a} \left(\left. \frac{\partial y}{\partial x} \right|_{\frac{j+1}{2}} - \left. \frac{\partial y}{\partial x} \right|_{\frac{j-1}{2}} \right) \quad (3.4)$$

$$= -\left. \frac{\partial^2 y}{\partial x^2} \right|_j \quad (3.5)$$

$$\Rightarrow \ddot{y} = \frac{T}{\mu} y'' \quad (3.6)$$

This equation is called *the wave equation*.

Definition 3.1.1 (The Wave Equation)

Let v be the phase velocity of a type of wave in a material or space. The wave equation is given by

$$\ddot{\psi} = v^2 \nabla^2 \psi \quad (3.7)$$

In the definition above, we have extended the domain to vector functions as well. The dispersion of this continuous string is thus

$$\omega^2 = \frac{T}{\mu} k^2 \Rightarrow \omega = \sqrt{\frac{T}{\mu}} k \quad (3.8)$$

It is possible to find two velocities associated with the dispersion; there is the *phase velocity* and the *group velocity*:

Definition 3.1.2 (Phase Velocity)

$$v_\varphi = \frac{\omega}{k} \quad (3.9)$$

Definition 3.1.3 (Group Velocity)

$$v_g = \frac{\partial \omega}{\partial k} \quad (3.10)$$

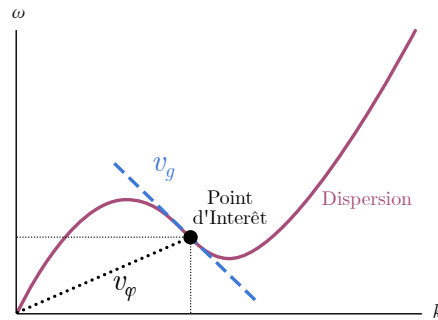


Figure 13. Dispersion showing the different velocities.

The phase velocity describes the propagation speed of a peak in time, and the group velocity describes the propagation speed of a group, of a waveform, in time. It is possible for the group velocity to be negative, as in the figure above. In the example of the string, the velocity v is thus $\sqrt{T/\mu}$.

We can also model compression waves in air, according to the following hypotheses:

- The fluid is perfect and compressible.
- The perturbations are weak (acoustic approximation).
- The motion is adiabatic and reversible.
- We neglect the effects of viscosity and thermal conduction.

The basic equations begin with conservation of mass:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (3.11)$$

where ρ is the mass density and $\mathbf{v}$ the fluid velocity. We also conserve momentum:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla p \quad (3.12)$$

where p is the pressure. For a reversible adiabatic transformation, we have that

$$p = C \rho^\gamma \quad (3.13)$$

where $\gamma = \frac{C_p}{C_v}$ is the Laplace coefficient (1.4 for air) and C a constant. The acoustic approximation allows us to consider only weak perturbations around an equilibrium state:

$$\rho = \rho_0 + \rho_1 \quad (\rho_1 \ll \rho_0) \quad (3.14)$$

$$p = p_0 + p_1 \quad (p_1 \ll p_0) \quad (3.15)$$

$$\mathbf{v} = \mathbf{v}_1 \quad (\mathbf{v}_1 \text{ small}) \quad (3.16)$$

Neglecting second-order terms,

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \nabla \cdot \mathbf{v}_1 = 0 \quad (3.17)$$

$$\rho_0 \frac{\partial \mathbf{v}_1}{\partial t} = -\nabla p_1 \quad (3.18)$$

Using $p = C \rho^\gamma$,

$$p_1 = \left(\frac{\partial p}{\partial \rho} \right) \bigg|_0 \rho_1 = c^2 \rho_1 \quad (3.19)$$

where $c = \sqrt{\gamma \frac{p_0}{\rho_0}}$ is the speed of sound. Taking the derivative of the continuity equation with respect to time and using Euler's equation, we obtain:

$$\frac{\partial^2 \rho_1}{\partial t^2} = -\rho_0 \nabla \cdot \left(\frac{\partial \mathbf{v}_1}{\partial t} \right) = \nabla \cdot (\nabla p_1) = \nabla^2 p_1 \quad (3.20)$$

Now, $p_1 = c^2 \rho_1$, so:

$$\frac{\partial^2 p_1}{\partial t^2} = c^2 \nabla^2 p_1 \quad (3.21)$$

We obtain the classical wave equation for the acoustic pressure p_1 :

$$\boxed{\ddot{\psi}_p = c^2 \nabla^2 \psi_p} \quad (3.22)$$

This equation describes the propagation of compression waves in air. We can also describe this phenomenon in terms of displacement, $\psi \propto \nabla \psi_p$:

$$\ddot{\psi} = c^2 \nabla^2 \psi \quad (3.23)$$

The equation is the same, but the distinction will emerge for boundary conditions.

3.2 Boundary Conditions

Until now we have considered only infinite systems – finite systems can be modeled as infinite systems but with some constraints that ensure correct modeling at the boundaries of the physical system.

3.2.1 Standing Waves

It is possible to create a *standing wave*, which does not transport energy:

$$\psi_{\text{stat}} = \psi_{\text{forward}} + \psi_{\text{backward}} = \text{Re} \left\{ \sum A_n (e^{ik_n x} + e^{-ik_n x}) e^{-i\omega_n t} \right\} = \sum A_n \sin(k_n x) \cos(\omega_n t) \quad (3.24)$$

In a simple case,

$$y = y_R + y_L = A (\sin(kx - \omega t) + \sin(kx + \omega t)) = A \sin(kx) \cos(\omega t) \quad (3.25)$$

Let us return to the string. Consider a string having length L , attached at both ends, as below.

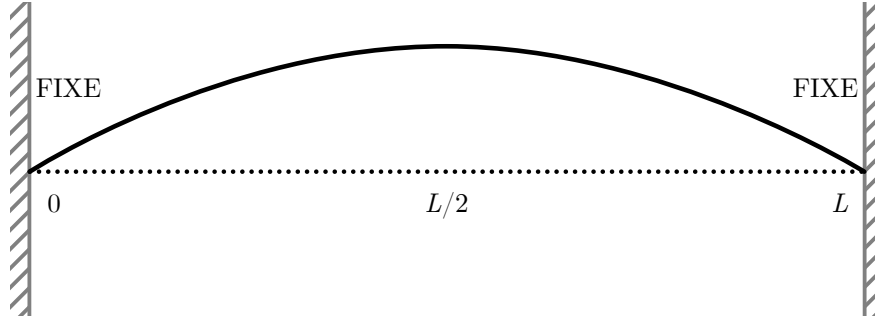


Figure 14. String with fixed ends.

It is also possible to have free ends:

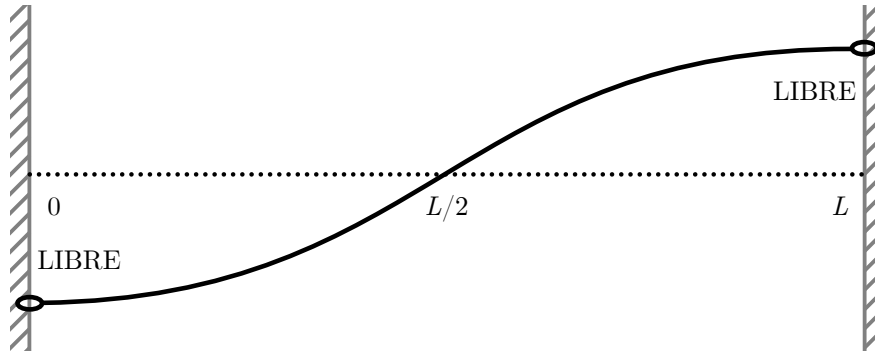


Figure 15. String with free ends.

Or one of each:

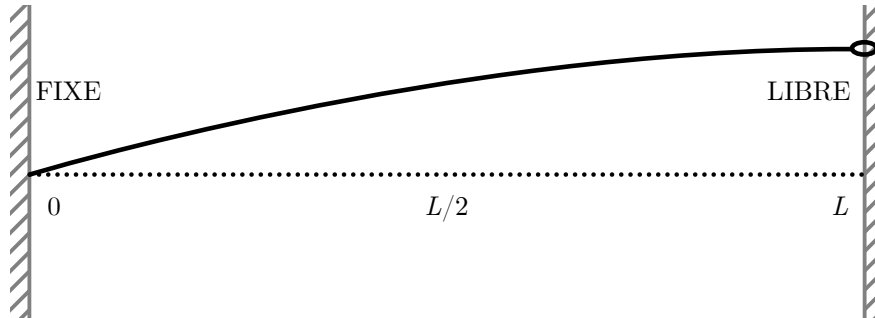


Figure 16. String with one free end and one fixed end.

What are the associated conditions? At a fixed end, the displacement $y(0)$ or $y(L)$ is fixed at zero; at a free end, no force is permitted and thus $y'(0)$ or $y'(L)$ is zero, as indicated in the sketches.

We have already established solutions having the form

$$y(x) = \sum A_n \cos(k_n x) \cos(\omega(k_n) t) + \sum B_n \sin(k_n x) \cos(\omega(k_n) t) \quad (3.26)$$

Suppose that the left end is fixed. Thus, all coefficients A_i would be zero; if the left end were free, all coefficients B_i would be zero.

The right end will decide the permitted k s. For sinusoidal modes (for the fixed left end), it is necessary that

$$\sin(kL) = 0 \Rightarrow k = \frac{n\pi}{L}, \quad n \in \mathbb{N} \quad (3.27)$$

For fixed-free ends, it is necessary that

$$\sin(kL) = \pm 1 \Rightarrow k = \frac{(2n-1)\pi}{2L}, \quad n \in \mathbb{N} \quad (3.28)$$

For free-free ends, k is also $n\pi/L$; it is possible to shift the constraints by $L/2$ to obtain the same constraints as for the fixed-fixed case.

3.2.2 Fourier Decomposition

Given an initial shape of a string, we can describe the resulting wave from the initial conditions. Usually, the temporal initial conditions are

$$\dot{y}(x) \Big|_{t=0} = 0 \quad ; \quad y(x) \Big|_{t=0} = y_0(x) \quad (3.29)$$

where y_0 is a function describing the initial shape. We can thus identify the coefficients A_n or B_n (depending on the condition at $x = 0$) using a Fourier series:

$$A_n = \frac{2}{L} \int_0^L y_0(x) \cos(k_n x) dx \quad ; \quad B_n = \frac{2}{L} \int_0^L y_0(x) \sin(k_n x) dx \quad (3.30)$$

Using these facts, one must know that the wave will decompose in time as a sum of two waves each propagating in one direction, as illustrated below.

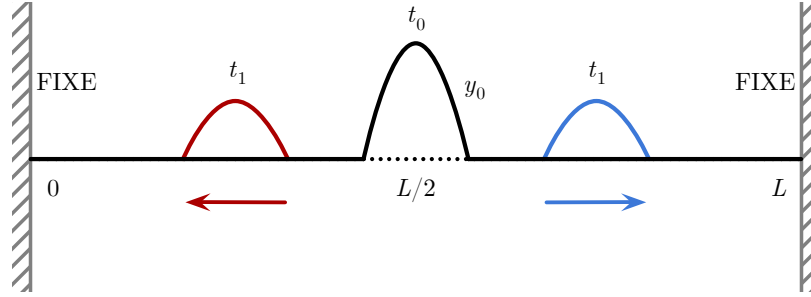


Figure 17. Decomposition of a wave.

But what happens when the wave reaches the boundary?

3.3 Impedances

We introduce the notion of *impedance*.

Definition 3.3.1 (Impedance)

Let q be a generalized coordinate oscillating in a wave-like manner. The impedance of a material with respect to a type of wave is defined as

$$Z = \frac{\text{force}}{\text{velocity}} = \frac{F}{\dot{q}} \quad (3.31)$$

For the string, the impedance of a string with tension T and linear density μ is

$$Z = \frac{-T y'}{\dot{y}} = \frac{-T(ik e^{ikx} e^{-i\omega t})}{-i\omega e^{ikx} e^{-i\omega t}} = T \frac{k}{\omega} = \frac{T}{v_\varphi} = \boxed{\sqrt{\mu T}} \quad (3.32)$$

3.3.1 Reflection and Transmission Coefficients

The concept of impedance leads directly to the concepts of *reflection and transmission coefficients*.

Proposition 3.3.2 (Reflection and Transmission Coefficients)

Let a material with impedance Z_1 have an interface with a material with impedance Z_2 such that one-dimensional waves can be transmitted from one medium to the other. The reflection and transmission coefficients are given by

$$r = \frac{Z_1 - Z_2}{Z_1 + Z_2} \quad ; \quad t = \frac{2Z_1}{Z_1 + Z_2} \quad (3.33)$$

Proof

Let a material with impedance Z_1 have an interface with a material with impedance Z_2 such that a wave ψ_0 leads to a reflection ψ_r and a transmission ψ_t . We require boundary conditions at this interface; namely, the string must be continuous (let $x = 0$ be the interface point):

$$\psi_0(0) + \psi_r(0) = \psi_t(0) \quad (3.34)$$

and the force must be zero:

$$T_1 \psi'_0(0) + T_1 \psi'_r(0) = T_2 \psi'_t(0) \quad (3.35)$$

For a general form

$$\psi = Ae^{ikx}e^{-i\omega t} \quad (3.36)$$

we obtain the equations below:

$$Ae^{-i\omega t} + rAe^{-i\omega t} = tAe^{-i\omega t} \Rightarrow 1 + r = t \quad (3.37)$$

and

$$T_1 k_1 A e^{-i\omega t} - r T_1 k_1 A e^{-i\omega t} = t T_2 k_2 A e^{-i\omega t} \Rightarrow \frac{T_1 k_1}{T_2 k_2} (1 - r) = t \quad (3.38)$$

For a shared ω , we have

$$\frac{k_1}{k_2} = \frac{v_1}{v_2} \quad (3.39)$$

and thus

$$\frac{T_1 k_1}{T_2 k_2} = \frac{Z_1}{Z_2} \quad (3.40)$$

Combining them,

$$1 + r = \frac{Z_1}{Z_2} - \frac{Z_1}{Z_2} r \Rightarrow \frac{Z_2 + Z_1}{Z_2} r = \frac{Z_1 - Z_2}{Z_2} \Rightarrow r = \frac{Z_1 - Z_2}{Z_1 + Z_2} \Rightarrow t = \frac{2Z_1}{Z_1 + Z_2} \quad (3.41)$$

as desired. $\square$

We observe that if the impedance of medium Z_2 is greater than that of medium Z_1 (thus $Z_2 > Z_1$), the reflection will accumulate a phase of π (and thus effectively a negative sign in front of the amplitude).

3.3.2 Successive Reflections

It is possible to have successive reflections, given three strings as below.

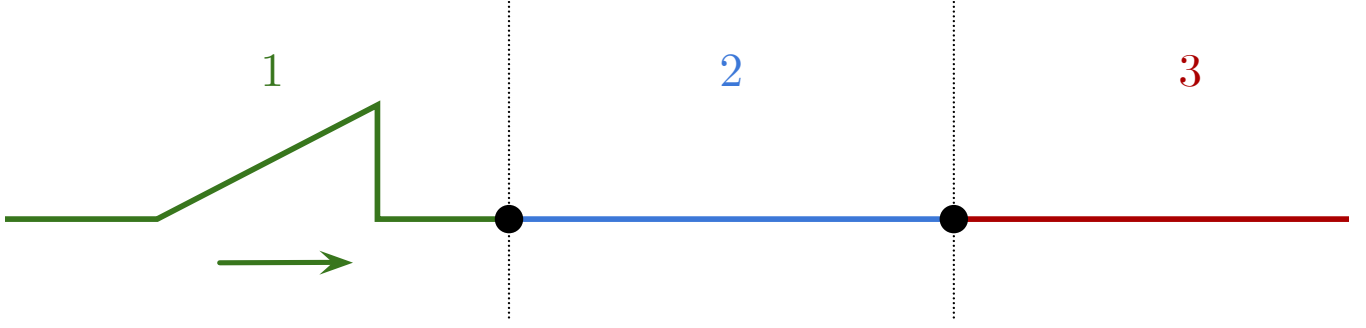


Figure 18. Three media with interfaces: initial situation.

Let a wave $\psi_0(x, t)$ propagate in medium 1 toward the right ($\mathbf{e}_1$). Let Z_1, Z_2, Z_3 be the impedances of the materials such that

$$Z_1 > Z_2 < Z_3 \quad (3.42)$$

Thus,

$$r_{12} = \frac{Z_1 - Z_2}{Z_1 + Z_2}; \quad t_{12} = \frac{2Z_1}{Z_1 + Z_2}; \quad r_{21} = \frac{Z_2 - Z_1}{Z_2 + Z_1}; \quad t_{21} = \frac{2Z_2}{Z_2 + Z_1}; \quad r_{23} = \frac{Z_2 - Z_3}{Z_2 + Z_3}; \quad t_{23} = \frac{2Z_2}{Z_2 + Z_3} \quad (3.43)$$

The waves would reflect in the following manner:

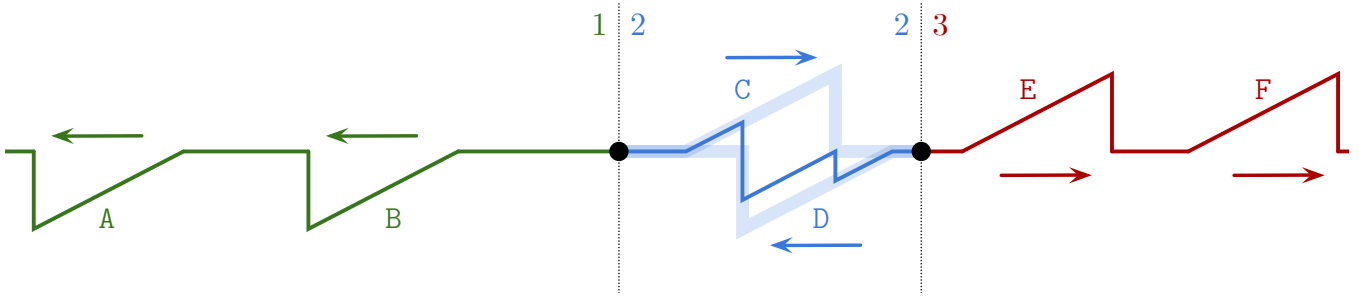


Figure 19. Three media with interfaces: situation after several internal reflections.

The wave that would have been created after the passage of wave ψ_0 through the boundary 1 – 2 will be

$$t_{12}\psi_0 \quad (3.44)$$

The first reflection, A in the diagram, is

$$r_{12}\psi_0 \quad (3.45)$$

After a transmission at 2 – 3, wave F is

$$t_{12}t_{23}\psi_0 \quad (3.46)$$

If we take the reflected wave at 2 – 3, its transmission into 1 will be wave B:

$$t_{12}r_{23}t_{21}\psi_0 \quad (3.47)$$

The reflection at 1 – 2 and transmission into 2 – 3 gives wave E:

$$t_{12}r_{23}r_{21}t_{23}\psi_0 \quad (3.48)$$

We thus observe that the n -th reflected wave in 1 will be

$$r_{12}\psi_0 \text{ for } n = 1; \quad t_{12}r_{23}(r_{21}r_{23})^{n-2}t_{21}\psi_0 \text{ for } n > 1 \quad (3.49)$$

In 3, the n -th transmitted wave will be

$$t_{12}(r_{23}r_{21})^{n-1}t_{23}\psi_0 \quad (3.50)$$

3.3.3 Power

The power associated with a wave $\psi(x, t)$ is given by

$$P = Z \left(\dot{\psi}(x, t) \right)^2 = ZA^2\omega^2 \exp(2i(kx - \omega t)) \quad (3.51)$$

The average power during a cycle of length $\Delta t \equiv 2\pi/\omega$ is

$$x = 0 : \langle P_0 \rangle = \frac{ZA^2\omega^2}{2} \quad (3.52)$$

3.4 Continuous Damping

The equation of a continuous but damped system would be

$$\mathbf{M}\ddot{\psi} + \mathbf{M}\Gamma\dot{\psi} + \mathbf{K}\psi = \mathbf{0} \quad (3.53)$$

for

$$\psi = \mathbf{A}_k e^{-i\omega t} \quad (3.54)$$

Thus, the eigenvalues are given by

$$[-\omega^2 - i\Gamma\omega + \mathbf{M}^{-1}\mathbf{K}] \mathbf{A}_k = \mathbf{0} \quad (3.55)$$

It is necessary that $\mathbf{A}_k$ be an eigenvector of $\mathbf{M}^{-1}\mathbf{K}$ and of Γ , thus

$$\mathbf{M}^{-1}\mathbf{K}\mathbf{A}_k = \omega_0^2(k)\mathbf{A}_k \quad ; \quad \Gamma\mathbf{A}_k = \gamma(k)\mathbf{A}_k \quad (3.56)$$

and thus the dispersion results immediately

$$\omega^2 = \omega_0^2(k) - i\gamma(k)\omega \quad (3.57)$$

If $\Gamma = \gamma_0 \mathbf{I}$, $\gamma(k)$ would not depend on k and would be equal to γ_0 . We see however that either k or ω (or both) must be complex for γ and ω_0 to remain real. Strange phenomena appear, such as the appearance of a standing wave near a driven end and a traveling wave far from the driven end (near a fixed end):

$$\tilde{\psi}(x, t) = A \left[\left(\frac{e^{i(k_r + ik_i)x} - e^{-i(k_r + ik_i)x}}{e^{i(k_r + ik_i)L} - e^{-i(k_r + ik_i)L}} \right) e^{-i\omega t} \right] \quad (3.58)$$

4 Waves in n Dimensions

4.1 Discrete Regime

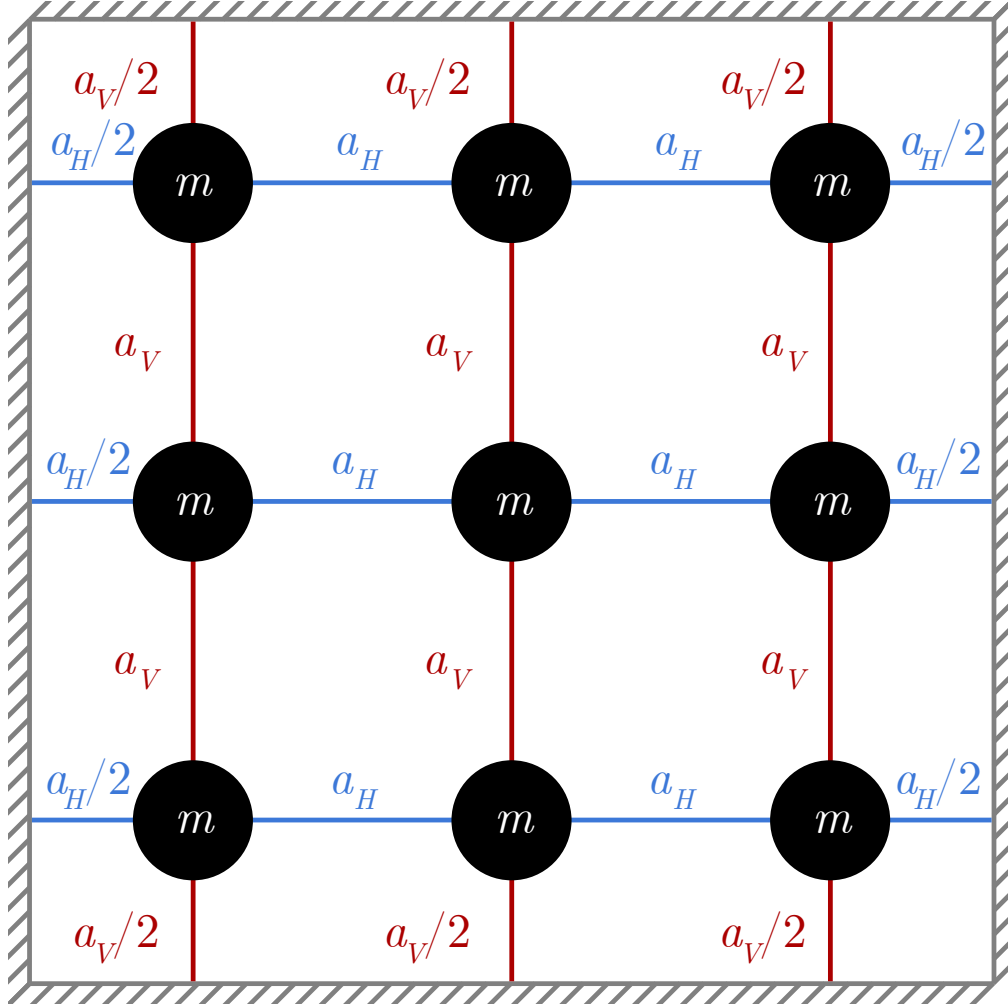


Figure 20. 2D discrete system.

Let a system of masses m be spaced a_H horizontally and a_V vertically. Let the tension of the vertical strings be T_V ; that of the horizontal ones T_H . Thus, the equation of motion becomes

$$\ddot{y} = -\frac{T_V}{ma_V}(2y_{i,j} - y_{i-1,j} - y_{i+1,j}) - \frac{T_H}{ma_H}(2y_{i,j} - y_{i,j-1} - y_{i,j+1}) \quad (4.1)$$

The dispersion thus becomes

$$\omega^2 = \frac{4T_V}{ma_V} \sin^2 \frac{k_V a_V}{2} + \frac{4T_H}{ma_H} \sin^2 \frac{k_H a_H}{2} \quad (4.2)$$

due to the linearity of the equations. The permitted modes for this system are

$$\langle 1, 1 \rangle ; \langle 1, 2 \rangle ; \langle 1, 3 \rangle ; \langle 2, 1 \rangle ; \langle 2, 2 \rangle ; \langle 2, 3 \rangle ; \langle 3, 1 \rangle ; \langle 3, 2 \rangle ; \langle 3, 3 \rangle \quad (4.3)$$

In general, for N rows and M columns, the modes are from $\langle 1, 1 \rangle$ to $\langle N, M \rangle$ in Cartesian product.

4.2 Continuous Regime

In the continuous regime, we make the same extrapolation for an orthotropic material:

$$\ddot{y} = \frac{T_V}{\mu_V} \frac{\partial y}{\partial x_V} + \frac{T_H}{\mu_H} \frac{\partial y}{\partial x_H} \Rightarrow \omega^2 = \frac{T_V}{\mu_V} k_v^2 + \frac{T_H}{\mu_H} k_H^2 \quad (4.4)$$

for $\mu_i = m/a_i$. For an isotropic 2D material, T and μ do not vary with direction, and thus

$$\ddot{y} = \frac{T}{\mu} \nabla^2 y \Rightarrow \omega^2 = \frac{T}{\mu} |\mathbf{k}|^2 \quad (4.5)$$

where $\mathbf{k} = \langle k_H, k_V \rangle$ is the *wave vector* such that

$$\mathbf{k} \cdot \mathbf{r} = \sum k_i x_i \mathbf{e}_i \quad (4.6)$$

and thus a *plane* wave would take the form

$$\psi = A e^{i\mathbf{k} \cdot \mathbf{r}} e^{-i\omega t} \quad (4.7)$$

In air, for example, we have

$$\psi \propto \nabla \psi_p \Rightarrow \omega^2 = v^2 \mathbf{k}^2 \quad (4.8)$$

In a box of size $\langle x_1, x_2, x_3 \rangle \in \{\mathbf{r} \in \mathbb{R}^3 \mid \mathbf{r} \cdot \mathbf{e}_1 \in [0, X_1]; \mathbf{r} \cdot \mathbf{e}_2 \in [0, X_2]; \mathbf{r} \cdot \mathbf{e}_3 \in [0, X_3]\}$, we have

$$\psi_p = A_p \prod \cos\left(\frac{n_i \pi x_i}{X_i}\right) \Rightarrow \omega^2 = v^2 \prod \left(\frac{n_i \pi}{X_i}\right)^2 \quad (4.9)$$

The displacement would be

$$\psi = A \prod \sin\left(\frac{n_i \pi x_i}{X_i}\right) \quad (4.10)$$

4.3 Chladni Plates

Let a square (side L) of isotropic 2D material (thus thin). We apply a transverse force $\mathbf{F}(t) = A \cos(\omega t) \mathbf{e}_3$ at the center. Let $\psi(x, y, t)$ be the transverse displacement of the material in the vertical axis. Thus, we require

$$\psi(0, 0, t) = A \cos(\omega t) \quad ; \quad \hat{\mathbf{n}} \cdot \nabla \psi = 0 \quad (4.11)$$

such that $\hat{\mathbf{n}}$ is either $\pm \mathbf{e}_1$ or $\pm \mathbf{e}_2$. Thus,

$$\frac{\partial}{\partial x} \psi(|L|, y, t) = \frac{\partial}{\partial y} \psi(x, |L|, t) = 0 \quad (4.12)$$

We thus expect modes having the form

$$\psi_{n_x, n_y}(x, y, t) = A \cos \frac{n_x \pi x}{L} \cos \frac{n_y \pi y}{L} \cos \omega t \quad (4.13)$$

with

$$\omega^2 = \omega_0^2(\mathbf{k}^2) = f(n_x^2 + n_y^2) \quad (4.14)$$

where f is a function $\mathbb{R} \rightarrow \mathbb{R}$. The dependence on $n_x^2 + n_y^2$ takes into account the rotational symmetry of the system, concretely of 90° . For a linear combination thus of mode $\langle a, b \rangle$ and mode $\langle b, a \rangle$, we see that their difference vanishes at the origin, and the sum remains:

$$\forall t : \psi_{n_x, n_y}^-(0, 0, t) = A \left(\cos \frac{n_x \pi x}{L} \cos \frac{n_y \pi y}{L} - \cos \frac{n_y \pi x}{L} \cos \frac{n_x \pi y}{L} \right) \cos \omega t = 0 \quad (4.15)$$

$$\exists t : \psi_{n_x, n_y}^+(0, 0, t) = A \left(\cos \frac{n_x \pi x}{L} \cos \frac{n_y \pi y}{L} + \cos \frac{n_y \pi x}{L} \cos \frac{n_x \pi y}{L} \right) \cos \omega t \neq 0 \quad (4.16)$$

Thus, the resonant modes of Chladni Plates are

$$\boxed{\psi_{n_x, n_y}(x, y, t) = A \left(\cos \frac{n_x \pi x}{L} \cos \frac{n_y \pi y}{L} + \cos \frac{n_y \pi x}{L} \cos \frac{n_x \pi y}{L} \right) \cos \omega t} \quad (4.17)$$

If we place sand on a Chladni plate, we can see the places where $\psi_{n_x, n_y}(x, y, t) = 0 \forall t$. Some of these modes are illustrated on the next page.

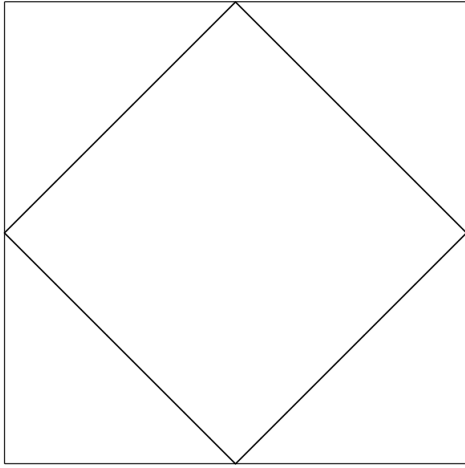


Figure 21. $\langle 1, 0 \rangle$. The lines are $y = \pm L \pm x$.

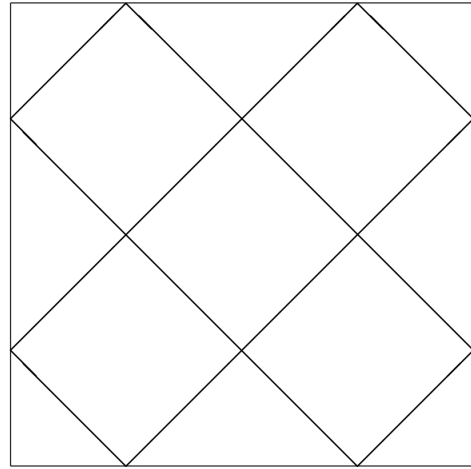


Figure 23. Mode $\langle 2, 0 \rangle$.

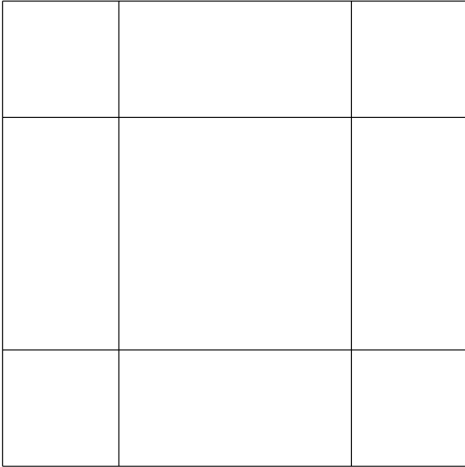


Figure 22. $\langle 1, 1 \rangle$. The lines are $y = \pm(L \pm L/2) \pm x$.

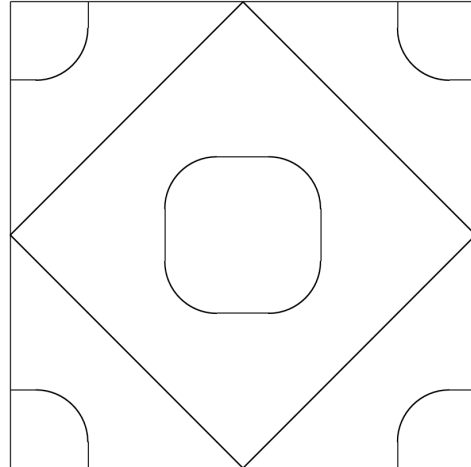


Figure 24. Mode $\langle 2, 1 \rangle$.

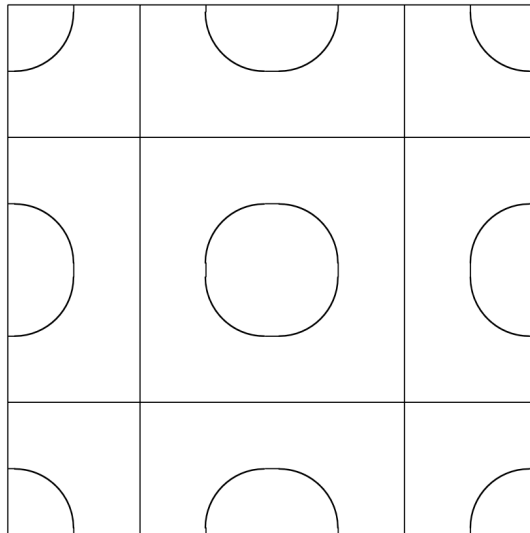


Figure 25. Mode $\langle 3, 1 \rangle$.

5 Electromagnetic Waves

5.1 The Wave Equation in EM

We begin from Maxwell's equations in vacuum:

$$\nabla \cdot \mathbf{E} = 0 \quad \nabla \cdot \mathbf{B} = 0 \quad (5.1)$$

$$\nabla \times \mathbf{E} = -\dot{\mathbf{B}} \quad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \dot{\mathbf{E}} \quad (5.2)$$

We can demonstrate that the wave equation applies: Let $\mathbf{B}' = c\mathbf{B}$:

$$\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}'}{\partial t} \quad \nabla \times \mathbf{B}' = \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \quad (5.3)$$

The curl becomes

$$\nabla \times (\nabla \times \mathbf{E}) = \cancel{\nabla(\nabla \cdot \mathbf{E})} - \nabla^2(\mathbf{E}) = \nabla \times \left(-\frac{1}{c} \frac{\partial \mathbf{B}'}{\partial t} \right) = -\frac{1}{c} \frac{\partial}{\partial t} (\nabla \times \mathbf{B}') = -\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (5.4)$$

$$\nabla \times (\nabla \times \mathbf{B}') = \cancel{\nabla(\nabla \cdot \mathbf{B}')} - \nabla^2(\mathbf{B}') = \nabla \times \left(\frac{1}{c} \frac{\partial \mathbf{E}}{\partial t} \right) = \frac{1}{c} \frac{\partial}{\partial t} (\nabla \times \mathbf{E}) = -\frac{1}{c^2} \frac{\partial^2 \mathbf{B}'}{\partial t^2} \quad (5.5)$$

Thus, we see that the wave equation applies to $\mathbf{E}$ and $\mathbf{B}'$:

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \nabla^2 \mathbf{B}' = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}'}{\partial t^2} \quad (5.6)$$

Because $\mathbf{B}' = c\mathbf{B}$, the differential equation does not change when we replace $\mathbf{B}'$ with $\mathbf{B}$:

$$\boxed{\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}} \quad \boxed{\nabla^2 \mathbf{B} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2}} \quad (5.7)$$

as desired. The dispersion is obviously

$$\omega^2 = c^2 k^2 \Rightarrow \omega = ck \quad (5.8)$$

We consider only plane waves for now, which will have the form

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{A} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (5.9)$$

Given an $\mathbf{E}$, the associated $\mathbf{B}$ is given by

$$\mathbf{B} = \frac{\mathbf{k} \times \mathbf{E}}{\omega} = \frac{n \mathbf{k} \times \mathbf{E}}{c} \quad (5.10)$$

The Poynting vector, $\mathbf{S}$, is defined as

$$\mathbf{S} = \frac{\mathbf{E} \times \mathbf{B}}{\mu_0} \quad (5.11)$$

5.1.1 Index of Refraction

The index of refraction of a material having permittivity ϵ (instead of ϵ_0) is given by

$$n = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}} \simeq \sqrt{\frac{\epsilon}{\epsilon_0}} = \frac{1}{nc\epsilon_0} \quad (5.12)$$

This gives

$$k = \sqrt{\frac{\mu \epsilon}{\mu_0 \epsilon_0}} \frac{\omega}{c} = n \frac{\omega}{c} \quad (5.13)$$

The energy density in an electromagnetic wave is given by

$$u = \frac{\epsilon \mathbf{E}^2}{2} + \frac{\mathbf{B}^2}{2\mu} \quad (5.14)$$

The momentum density is given by

$$\mathbf{p} = \epsilon \mathbf{E} \times \mathbf{B} \quad (5.15)$$

Thus,

$$\mathbf{S} = c^2 \mathbf{p} \quad (5.16)$$

Clearly,

$$\dot{u} + \nabla \cdot \mathbf{S} = 0 \quad (5.17)$$

The time average of the Poynting vector is calculated as below:

$$\langle \mathbf{S} \rangle = \hat{\mathbf{k}} \sqrt{\frac{\epsilon}{\mu}} \langle u \rangle \quad (5.18)$$

where

$$\langle u \rangle = \frac{1}{T} \int_T u(t) dt = \frac{1}{T} \int_T \frac{\epsilon}{4\pi} A_i \cos^2(\mathbf{k} \cdot \mathbf{r} - \omega t + \varphi_i) dt = \frac{\epsilon}{8\pi} A_i^2 \quad (5.19)$$

One must also define the intensity:

$$I = \frac{\text{power}}{\text{area}} = \frac{\dot{u}}{A} \propto \frac{\langle |\mathbf{S}| \rangle}{A} \propto \frac{E^2}{A} \quad (5.20)$$

Thus, the intensity of a superposition of electromagnetic waves with magnitudes $\{\dots E_i \dots\}$ and thus intensities $\{\dots I_i \dots\}$ is

$$I_{\text{tot}} = \left(\sum_i \sqrt{I_i} \right)^2 \quad (5.21)$$

Thus, for two waves,

$$I_{\text{tot}} = \left(\sqrt{I_0} + \sqrt{I_0} \right)^2 = 4I_0 \quad (5.22)$$

5.2 Snell's Law

Given this, we calculate the interaction of a plane wave with a boundary between two dielectrics having indices of refraction n and n' . Let the boundary be the plane $z = 0$. Let the incoming wave be polarized in the x axis (thus $\mathbf{e}_1$)

$$\mathbf{E}_0 = A \mathbf{e}_1 e^{i(kz - \omega t)} \Rightarrow \mathbf{E}_R = r A \mathbf{e}_1 e^{i(-kz - \omega t)} \quad ; \quad \mathbf{E}_T = t A \mathbf{e}_1 e^{i(k'z - \omega t)} \quad (5.23)$$

and

$$\mathbf{B}_0 = \frac{n}{c} A \mathbf{e}_1 e^{i(kz - \omega t)} \Rightarrow \mathbf{B}_R = -r \frac{n}{c} A \mathbf{e}_1 e^{i(-kz - \omega t)} \quad ; \quad \mathbf{B}_T = t \frac{n'}{c} A \mathbf{e}_1 e^{i(k'z - \omega t)} \quad (5.24)$$

The boundary conditions are

$$\Delta \mathbf{E} = 0 \quad ; \quad \Delta \mathbf{B} = 0 \quad (5.25)$$

Thus,

$$1 + r = t \quad ; \quad n(1 - r) = n't \Rightarrow \boxed{t = \frac{2}{1 + n'/n}} \quad ; \quad \boxed{r = \frac{n - n'}{n + n'}} \quad (5.26)$$

Suppose we have a general plane wave

$$\psi(\mathbf{r}, t) = A e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (5.27)$$

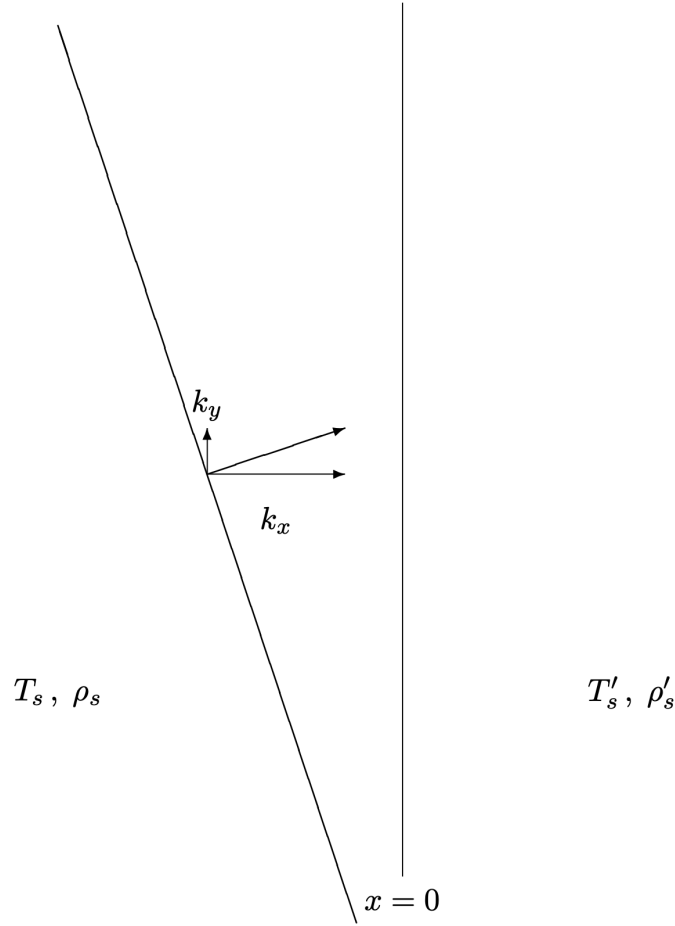


Figure 26. A plane wave at a translationally invariant boundary.

For translational invariance to apply all along the plane $x = 0$, it is necessary that k_y not change on either side of the boundary. However, k_x can of course change: the reflection will have $-k_x$ and the transmission will have k'_x . Thus,

$$k_x = \sqrt{\frac{\omega^2}{v^2} - k_y^2} \quad ; \quad k'_x = \sqrt{\frac{\omega^2}{(v')^2} - k_y^2} \quad (5.28)$$

with $v/v' = n'/n$. Thus, for θ and θ' angles with the normal,

$$\sin \theta = \frac{k_y}{k_x} \quad ; \quad \sin \theta' = \frac{k_y}{k'_x} \Rightarrow \frac{\sin \theta}{\sin \theta'} = \frac{k'_x}{k_x} = \frac{n'}{n} \quad (5.29)$$

Proposition 5.2.1 (Snell's Law)

Let an interface exist between two materials with indices of refraction n, n' . The angles of incidence are related by

$$n \sin \theta = n' \sin \theta' \quad (5.30)$$

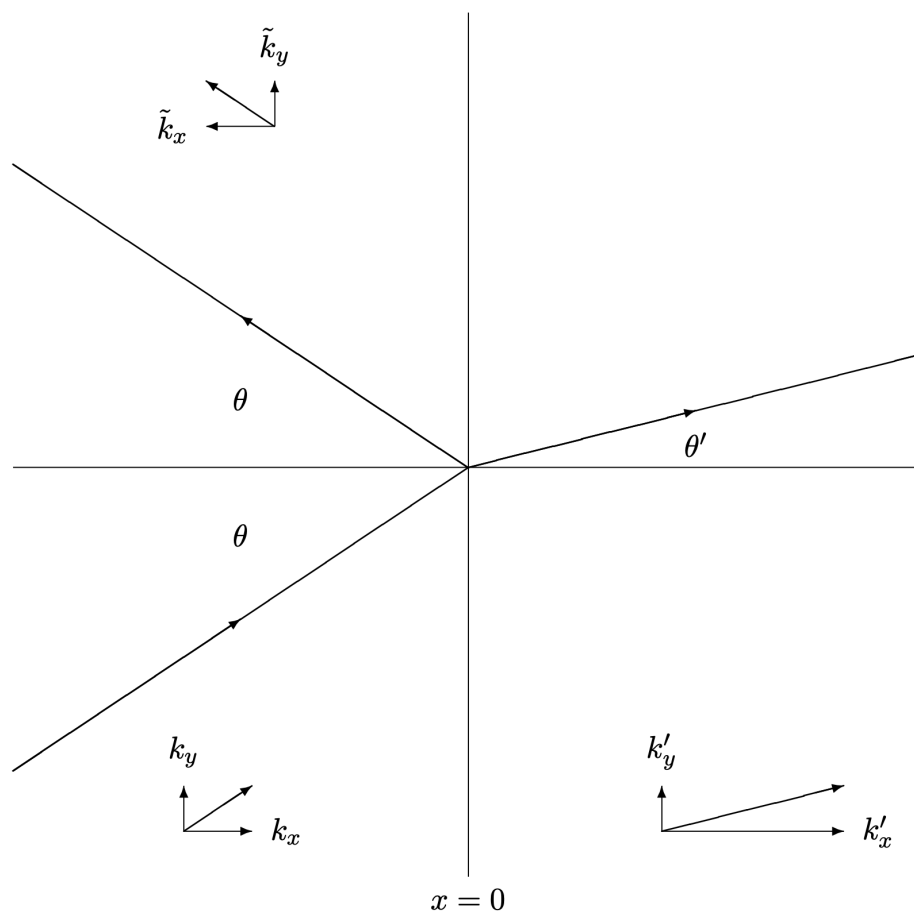


Figure 27. Snell's Law, illustrated.

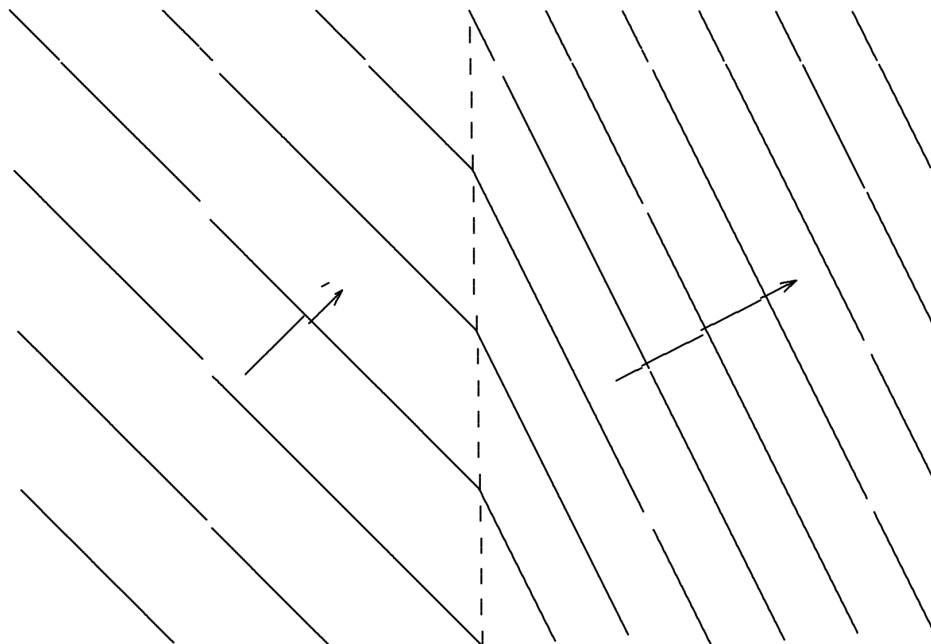


Figure 28. Refraction (reflection omitted).

It is possible to find a combination θ , n , and n' such that $\theta' = \pi/2$:

$$n \sin \theta = n' \sin \theta' = n' \Rightarrow \boxed{\theta_{\text{crit}} = \arcsin \frac{n'}{n}} \quad (5.31)$$

Any angle of incidence greater than or equal to this θ_{crit} would lead to a total lack of transmission, which is called *total internal reflection*. Effectively, k'_x becomes imaginary.

5.3 Polarization

The *polarization* of an electromagnetic wave $\mathbf{E}(\mathbf{r}, t)$ is given by

$$\mathbf{Z} : \mathbf{E}(\mathbf{r}, t) = \mathbf{Z} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (5.32)$$

There are three main families of polarizations. A polarization having the form

$$\mathbf{Z} = \begin{bmatrix} a \\ b \end{bmatrix} ; a, b \in \mathbb{R} \quad (5.33)$$

is “linear” – the electric field is aligned along the axis defined by $\langle a, b \rangle$. A polarization having the form

$$\mathbf{Z} = a \begin{bmatrix} 1 \\ i \end{bmatrix} ; a \in \mathbb{R} \quad (5.34)$$

is “circular”: effectively one component is phase-shifted by $\pi/2$ (thus receives a factor of i in front). A polarization having the form

$$\mathbf{Z} = \begin{bmatrix} a \\ bi \end{bmatrix} ; a, b \in \mathbb{R} \quad (5.35)$$

is “elliptical”: the $\mathbf{e}_1$ and $\mathbf{e}_2$ radii are not equal.

We see from equations describing the energy of an EM wave that

$$\langle \mathbf{S} \rangle \propto |\mathbf{Z}|^2 \quad (5.36)$$

5.3.1 Polarizers

A *polarizer* allows a wave with multiple polarizations to become polarized in a particular direction. A “comb” having metal bars (vertical, say) will allow horizontally polarized waves (thus $\mathbf{Z} = \mathbf{e}_1$) to pass but not others. This kind of “filter” is very inefficient, because all the energy associated with non-permitted polarizations is lost.

This is represented in matrix form, using the projection operator:

$$\mathbf{P}_\theta = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \quad (5.37)$$

such that

$$\mathbf{P}_\theta \mathbf{P}_\theta = \mathbf{P}_\theta \quad (5.38)$$

This matrix takes any vector $\mathbf{Z}$ and projects it onto the axis $u_\theta \equiv \langle \cos \theta, \sin \theta \rangle$ (the eigenvector with eigenvalue $\lambda = 1$):

$$\mathbf{P}_\theta \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \begin{bmatrix} \cos^3 \theta + \sin^2 \theta \cos \theta \\ \sin^3 \theta + \sin \theta \cos^2 \theta \end{bmatrix} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \quad (5.39)$$

We also see that the orthogonal vector, $\langle -\sin \theta, \cos \theta \rangle$, is the eigenvector with eigenvalue $\lambda = 0$ (thus these polarizations are completely attenuated):

$$\mathbf{P}_\theta \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix} = \mathbf{0} \quad (5.40)$$

5.3.2 Quarter-Wave Plates

It is also possible to use the property of birefringence to create a thin filter that will add a phase φ to one component but not to another. The kind of plate that introduces a difference of $\pi/2$ is called a *quarter-wave plate*; we can also write a matrix for this phenomenon, such that

$$\mathbf{Q}_\theta = \mathbf{P}_\theta + i\mathbf{P}_{\theta+\pi/2} = \begin{bmatrix} \cos^2 \theta + i \sin^2 \theta & (1-i) \sin \theta \cos \theta \\ (1-i) \sin \theta \cos \theta & \sin^2 \theta + i \cos^2 \theta \end{bmatrix} \quad (5.41)$$

A *half-wave plate* (introducing a difference of π) is defined in a similar manner:

$$\mathbf{H}_\theta = \mathbf{P}_\theta - \mathbf{P}_{\theta+\pi/2} = \mathbf{Q}_\theta \mathbf{Q}_\theta = \begin{bmatrix} (\cos^2 \theta + i \sin^2 \theta)^2 & -2i \sin^2 \theta \cos^2 \theta \\ -2i \sin^2 \theta \cos^2 \theta & (i \cos^2 \theta + \sin^2 \theta)^2 \end{bmatrix} \quad (5.42)$$

Certain properties become apparent:

$$\mathbf{Q}_0 \mathbf{P}_{\pi/4} \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} = \frac{\psi_1 + \psi_2}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} \quad (5.43)$$

In words, a wave with linear polarization diagonal compared to a quarter-wave plate (thus $\theta_{\mathbf{Z}} = \pi/4$ for a plate $\mathbf{Q}_0$) will receive circular polarization after the quarter-wave plate.

5.4 Birefringence

Let a material, like CaCO_3 , be such that the index of refraction depends on the axes. Suppose that for the x axis it is n_x , and for y , n_y . Suppose there is a wave with wavelength λ such that

$$\mathbf{Z} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (5.44)$$

The phase accumulated in traversing a distance d is

$$\varphi_i = k_i d \quad (5.45)$$

In a material with index of refraction n , $k = n\omega/c$ and thus

$$\varphi_i = \frac{n_i \omega}{c} d \quad (5.46)$$

If we have rays that traverse the material but with different indices of refraction, we obtain a phase shift between the two polarizations ($\mathbf{Z}_1 = \mathbf{e}_1$; $\mathbf{Z}_2 = \mathbf{e}_2$):

$$\Delta\varphi = \frac{\Delta n}{c} \omega d \quad (5.47)$$

Thus, the thickness to obtain a quarter-wave plate is

$$d = \frac{c}{\omega \Delta n} \cdot \left(\frac{\pi}{2} + 2m\pi \right) \quad (5.48)$$

If we wanted to attenuate the wave entirely,

$$d = \frac{c}{\omega \Delta n} \cdot (\pi + 2m\pi) \quad (5.49)$$

If we enter at an angle a with respect to the coordinate system of the indices of refraction, and we take the approximation that the rays polarized in $\mathbf{e}_1$ and $\mathbf{e}_2$ do not separate much, the distances change as drawn below. Effectively,

$$n'_x = \frac{n_x}{\cos b} \quad ; \quad n'_y = \frac{n_y}{\cos c} \quad \implies \quad \Delta n_{\text{eff}} = \left| \frac{n_x}{\cos b} - \frac{n_y}{\cos c} \right| \quad (5.50)$$

for $b = \arcsin\left(\frac{\sin a}{n_x}\right)$ and $c = \arcsin\left(\frac{\sin a}{n_y}\right)$.

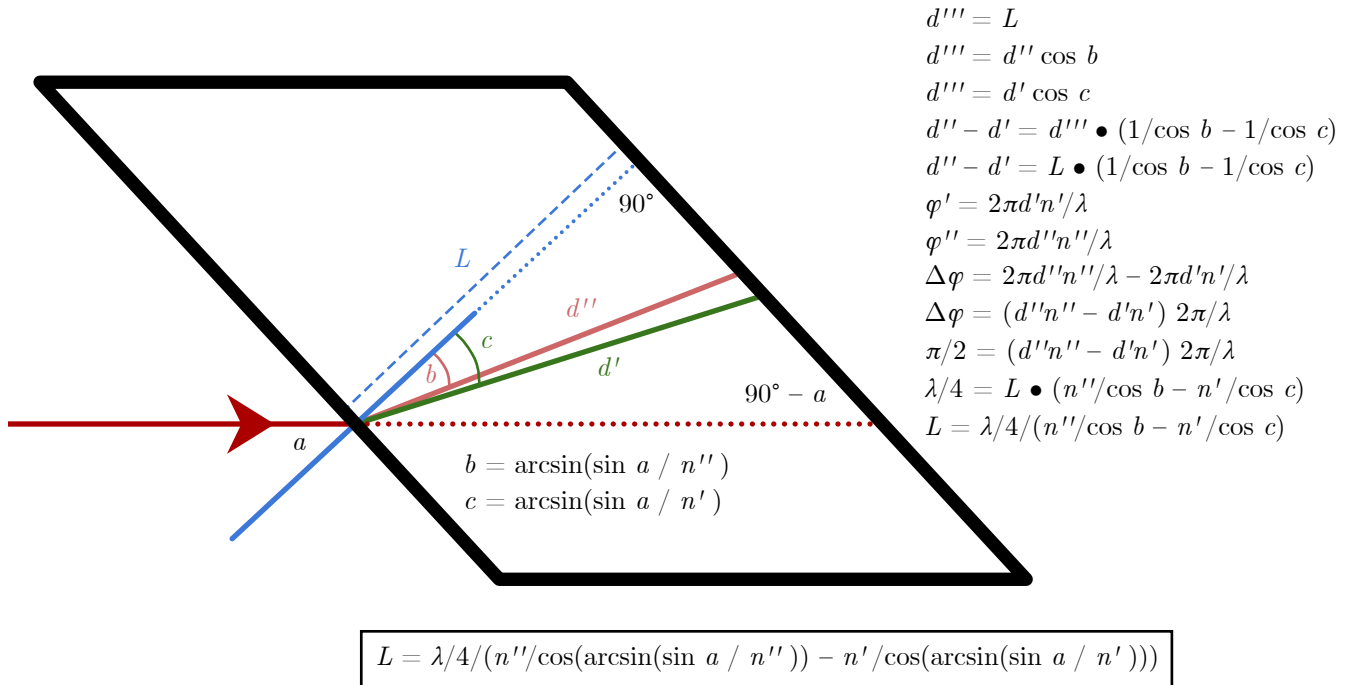


Figure 29. Birefringence at an angle.

5.5 P and S Polarizations

Let a planar interface surface exist between a region with index of refraction n_1 and a region with index of refraction n_2 . The S polarization is that normal to the interface; the P polarization is that parallel to the interface. The **power** reflection and **power** transmission coefficients for these polarizations are given by

$$R_S = \left| \frac{Z_2 \cos \theta_0 - Z_1 \cos \theta_t}{Z_2 \cos \theta_0 + Z_1 \cos \theta_t} \right|^2 \quad ; \quad T_S = 1 - R_S \quad (5.51)$$

$$R_P = \left| \frac{Z_2 \cos \theta_t - Z_1 \cos \theta_0}{Z_2 \cos \theta_t + Z_1 \cos \theta_0} \right|^2 \quad ; \quad T_P = 1 - R_P \quad (5.52)$$

where Z is the impedance of the material,

$$Z \equiv \sqrt{\frac{\mu}{\epsilon}} = \frac{1}{nc\epsilon_0} \quad (5.53)$$

5.5.1 Reflectance, Transmittance of Intensity

We recall that

$$\frac{|E_R|}{|E_I|} \equiv r = \frac{n - n'}{n + n'} \quad ; \quad \frac{|E_T|}{|E_I|} \equiv t = \frac{2n}{n + n'} \quad (5.54)$$

The power reflectance is given by

$$R = \frac{P_R}{P_I} = \frac{n_1 \epsilon_0 c}{2} |E_R|^2 A_R \cdot \left(\frac{n_1 \epsilon_0 c}{2} |E_I|^2 A_I \right)^{-1} = \frac{|E_R|^2}{|E_I|^2} = r^2 \quad (5.55)$$

The power transmittance is given by

$$T = \frac{P_T}{P_I} = \frac{n_2 \epsilon_0 c}{2} |E_T|^2 A_T \cdot \left(\frac{n_1 \epsilon_0 c}{2} |E_I|^2 A_I \right)^{-1} = \frac{n_2 A_T |E_T|^2}{n_1 A_I |E_I|^2} = \frac{n_2 \cos \theta_T}{n_1 \cos \theta_I} t^2 \quad (5.56)$$

We observe that $T + R = 1$, and not $1 + R = T$.

5.6 Brewster's Angle

Brewster's Angle allows the reflection of only one polarization.

Proposition 5.6.1 (Brewster's Angle)

Brewster's angle is the angle of incidence of an unpolarized wave from a region having index of refraction n_1 on the boundary of a region having index of refraction n_2 such that the polarization of the reflected wave is parallel to the interface. This angle is given by

$$\theta_B = \arctan \frac{n_2}{n_1} \quad (5.57)$$

Proof

We seek the angle θ_B such that $R_P = 0$:

$$\left| \frac{Z_2 \cos \theta_2 - Z_1 \cos \theta_B}{Z_2 \cos \theta_2 + Z_1 \cos \theta_B} \right|^2 = 0 \quad (5.58)$$

This happens when

$$Z_2 \cos \theta_2 = Z_1 \cos \theta_B \quad (5.59)$$

From Eq. (5.53), we see that

$$\frac{Z_2}{Z_1} = \frac{n_1}{n_2} \quad (5.60)$$

and thus

$$n_2 \cos \theta_B = n_1 \cos \theta_2 \Rightarrow n_2^4 \cos^2 \theta_B = n_1^2 n_2^2 \cos^2 \theta_2 \quad (5.61)$$

But we also have Snell's Law:

$$n_1 \sin \theta_B = n_2 \sin \theta_2 \Rightarrow n_1^4 \sin^2 \theta_B = n_1^2 n_2^2 \sin^2 \theta_2 \quad (5.62)$$

Let us add the equations:

$$n_1^4 \sin^2 \theta_B + n_2^4 \cos^2 \theta_B = n_1^2 n_2^2 (\sin^2 \theta_2 + \cos^2 \theta_2) = n_1^2 n_2^2 = n_1^2 n_2^2 (\sin^2 \theta_B + \cos^2 \theta_B) \quad (5.63)$$

$$\Rightarrow \tan^2 \theta_B + \frac{n_2^4}{n_1^4} = \frac{n_2^2}{n_1^2} \tan^2 \theta_B + \frac{n_2^2}{n_1^2} \quad (5.64)$$

$$\Rightarrow \tan^2 \theta_B \left(1 - \frac{n_2^2}{n_1^2} \right) = \frac{n_2^2}{n_1^2} \left(1 - \frac{n_2^2}{n_1^2} \right) \quad (5.65)$$

$$\Rightarrow \theta_B = \arctan \frac{n_2}{n_1} \quad (5.66)$$

as desired. $\square$

6 Radiation

6.1 Acceleration of a Charge

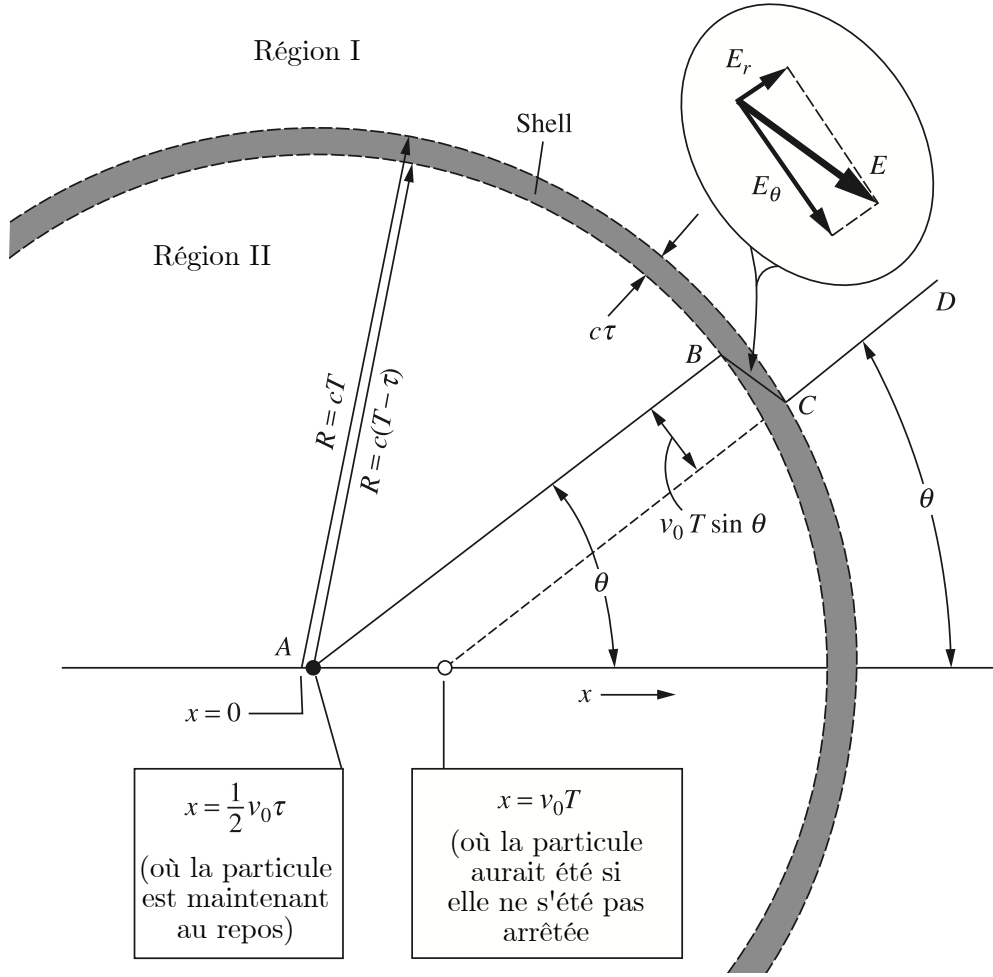


Figure 30. Acceleration of a charge. $T \gg \tau$

Consider a charge q that stops abruptly – before, it was moving with velocity $\mathbf{v} = v\mathbf{e}_1$, and in a time τ it stops completely. The electric field is drawn in the sketch above. We see that

$$\frac{E_\theta}{E_r} = \frac{v_0 T \sin \theta}{c\tau} \quad (6.1)$$

However, E_r cannot vary in time, due to Gauss's Law. Thus,

$$E_r = \frac{q}{4\pi\epsilon_0 R^2} \Rightarrow E_\theta = \frac{qv \sin \theta}{4\pi\epsilon_0 c^3 T \tau} \quad (6.2)$$

We know that v_0/τ is the acceleration a , and $cT = R$, and thus

$$E_\theta = \frac{qa \sin \theta}{4\pi\epsilon_0 c^2 R} \quad (6.3)$$

Thus, the field at a point $\mathbf{r}$ (for the charge q at the origin) will be

$$\mathbf{E}(\mathbf{r}, t) = \frac{\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{a}(t - |\mathbf{r}|/c))}{4\pi\epsilon_0 c^2} \cdot \frac{q}{r^2} \quad (6.4)$$

The associated magnetic field $\mathbf{B}$ is

$$\mathbf{B} = \frac{\mathbf{r} \times \mathbf{E}}{c} = \frac{\mathbf{k} \times \mathbf{E}}{c} \quad (6.5)$$

The energy in a spherical shell is

$$U_E = \iiint_{\text{shell}} u_E d\tau = \iiint_{\text{shell}} \frac{\epsilon_0 E_0 \theta^2}{2} d\tau = 4\pi R^2 c\tau \cdot \frac{q^2 a^2 \sin^2 \theta}{32\pi^2 \epsilon_0 R^2 c^4} \iint_{\text{surface}} \sin^2 \theta dA = \frac{q^2 a^2 \tau}{12\pi \epsilon_0 c^3} \quad (6.6)$$

The total energy will be double (due to the magnetic field):

$$U = U_E + U_B = \frac{q^2 a^2 \tau}{6\pi \epsilon_0 c^3} \quad (6.7)$$

Thus, the power (energy per time) is

$$P = \dot{U} = \frac{q^2 a^2}{6\pi \epsilon_0 c^3} \quad (6.8)$$

This is also called the Larmor formula.

6.2 The Dipole Antenna

A dipole antenna is a device that allows charges, in total Q , to oscillate sinusoidally:

$$\mathbf{r}_0(t) = l \mathbf{e}_3 \cos \omega t \quad (6.9)$$

Thus,

$$\mathbf{a}(t) = \ddot{\mathbf{r}}_0(t) = -l\omega^2 \mathbf{e}_3 \cos \omega t \quad (6.10)$$

Thus, the associated electric field is

$$\mathbf{E}(\mathbf{r}, t) = \frac{\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \mathbf{a}(t - |\mathbf{r}|/c))}{4\pi \epsilon_0 c^2} \cdot \frac{q}{\mathbf{r}^2} = \frac{q}{4\pi \epsilon_0 c^2 \mathbf{r}^2} \cos \left(\omega \left(t - \frac{|\mathbf{r}|}{c} \right) \right) (\mathbf{e}_3 - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{e}_3)) \quad (6.11)$$

Let θ be the angle from the $\mathbf{e}_3$ axis. Thus,

$$(\mathbf{e}_3 - \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{e}_3)) = \sin \theta \quad (6.12)$$

and thus

$$\mathbf{E}(\mathbf{r}, t) = \frac{q \sin \theta}{4\pi \epsilon_0 c^2 \mathbf{r}^2} \cos \left(\omega \left(t - \frac{|\mathbf{r}|}{c} \right) \right) \quad (6.13)$$

The intensity is

$$I(\mathbf{r}, t) = \frac{\epsilon_0 |\mathbf{E}(\mathbf{r}, t)|^2}{2} \cdot (4\pi \mathbf{r}^2) = \frac{q^2 \sin^2 \theta}{8\pi \epsilon_0 c^4 \mathbf{r}^2} \cos^2 \left(\omega \left(t - \frac{|\mathbf{r}|}{c} \right) \right) \quad (6.14)$$

On average (integrated over one cycle),

$$\langle I(\mathbf{r}) \rangle = \frac{q^2 \sin^2 \theta}{16\pi \epsilon_0 c^4 \mathbf{r}^2} \quad (6.15)$$

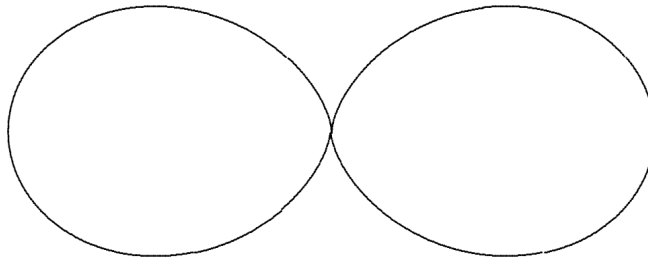


Figure 31. Radiation intensity of an antenna. The $\mathbf{e}_3$ axis is vertical and centered in the image.

6.3 Antenna Interference

It is of course possible to place multiple antennas near each other – if the frequencies are the same, we can observe interference of the intensities. If the frequencies are not the same, there will be no interference (because the average over a common period will be 0):

$$I_1 = I_0 \cos^2(\omega_1 t') \quad ; \quad I_2 = I_0 \cos^2(\omega_2 t') \quad (6.16)$$

We superpose the magnitudes of the fields:

$$E_1 = E_0 \cos(\omega_1 t' - k_1 r + \varphi_1) \quad ; \quad E_2 = E_0 \cos(\omega_2 t' - k_2 r + \varphi_2) \quad (6.17)$$

and thus

$$E_{12} = E_0 (\cos(\omega_1 t' - k_1 r + \varphi_1) + \cos(\omega_2 t' - k_2 r + \varphi_2)) \quad (6.18)$$

The combined intensity will be

$$I_{12} = |E_{12}|^2 \quad (6.19)$$

$$= E_0^2 (\cos(\omega_1 t' - k_1 r + \varphi_1) + \cos(\omega_2 t' - k_2 r + \varphi_2))^2 \quad (6.20)$$

$$= E_0^2 (\cos^2(\omega_1 t' - k_1 r + \varphi_1) + \cos^2(\omega_2 t' - k_2 r + \varphi_2) + 2 \cos(\omega_1 t' - k_1 r + \varphi_1) \cos(\omega_2 t' - k_2 r + \varphi_2)) \quad (6.21)$$

$$= E_0^2 (\cos^2(\omega_1 t' - k_1 r + \varphi_1) + \cos^2(\omega_2 t' - k_2 r + \varphi_2) + \frac{\cos((\omega_1 + \omega_2)t' - 2(k_1 + k_2)r + \varphi_1 + \varphi_2) + \cos((\omega_1 - \omega_2)t' - 2(k_1 - k_2)r + \varphi_1 - \varphi_2)}{2}) \quad (6.22)$$

We see that each term has a temporal dependence except the last one if $k_1 = k_2 \iff \omega_1 = \omega_2$, which would lead to

$$\langle I_{12} \rangle = cn\epsilon_0 E_0^2 (1 + \cos \Delta\varphi) \quad (6.23)$$

(and thus if $\omega_1 \neq \omega_2$, the middle term would vanish and thus there would be no spatial dependence of the intensity, and thus no interference.) In complex form, we generalize to

$$I = \left(\sum_i \sqrt{I_i} \right)^2 |1 + e^{i\Delta\varphi}|^2 \quad (6.24)$$

Let two antennas be a distance d apart. What is the accumulated phase difference?

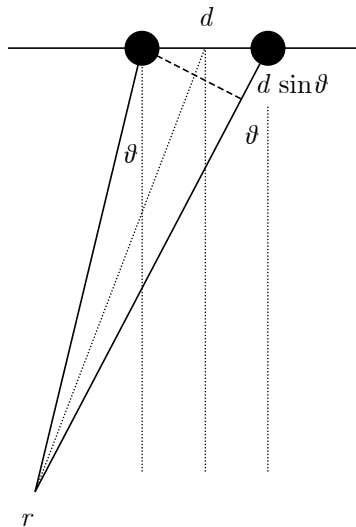


Figure 32. Pair of dipole antennas.

If we assume that $r \gg d$, we can approximate the angles between the vertical and the ray from each antenna as being equal. Thus, the longer path is longer by $d \sin \vartheta$ approximately, and thus the accumulated phase is

$$\Delta\varphi = kd \sin \vartheta = \frac{2\pi d \sin \vartheta}{\lambda} \quad (6.25)$$

Let $\Delta\varphi_0$ be an initial phase difference (unrelated to distances). Thus, the intensity as a function of this angle ϑ will be

$$I(\vartheta) = \left(\sum_i \sqrt{I_i} \right)^2 \cos^2 \left(\frac{\Delta\varphi_0}{2} + \frac{\pi}{\lambda} d \sin \vartheta \right) \quad (6.26)$$

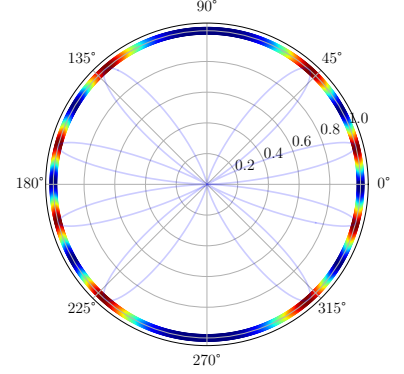
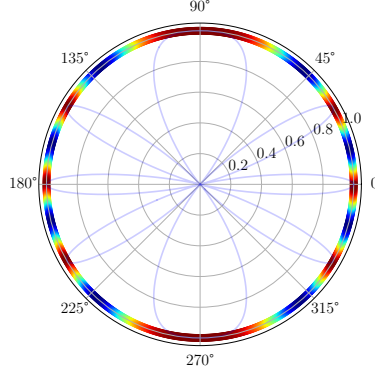
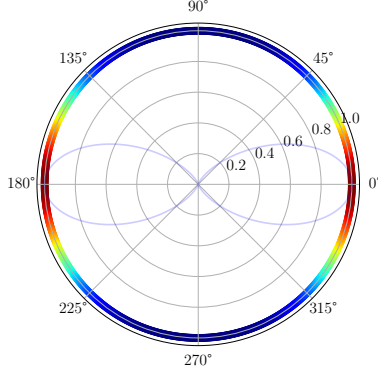


Figure 33. $I(\theta)$: $d = \lambda/2$, $\Delta\varphi_0 = 0$. **Figure 34.** $I(\theta)$: $d = 2\lambda$, $\Delta\varphi_0 = 0$. **Figure 35.** $I(\theta)$: $d = 2\lambda$, $\Delta\varphi_0 = \frac{\pi}{2}$.

7 Interference and Diffraction

In this final section, we explore basic optics: interference from multiple slits, diffraction effects, and applications to quantum mechanics.

7.1 Interference by Reflection

The simplest case is found in interference due to a point light source and a perfect mirror behind it, as below.

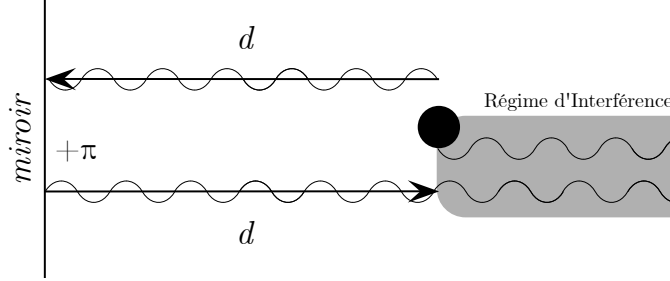


Figure 36. Mirror with point light source.

For a distance d from the mirror, the path of light that bounces off the mirror has length $2d$ with a phase of π added by the mirror such that

$$\Delta\varphi = 2kd + \pi \quad (7.1)$$

Thus, the relative phase between a wave that leaves the source and one that has bounced is given above; let m be a positive integer; the rays *interfere* to either cancel (when $\Delta\varphi = (2m \pm 1)\pi$) or add (when $\Delta\varphi = 2m\pi$):

$$\text{Constructive: } 2kd + \pi = 2m\pi \Rightarrow d = \frac{2m \pm 1}{2k}\pi = \boxed{\frac{2m \pm 1}{4}\lambda} \quad (7.2)$$

$$\text{Destructive: } 2kd + \pi = (2m + 1)\pi \Rightarrow d = \frac{m}{k}\pi = \boxed{\frac{m}{2}\lambda} \quad (7.3)$$

7.2 Thin Film Interference

We generalize: suppose we have a thin film, as drawn below:

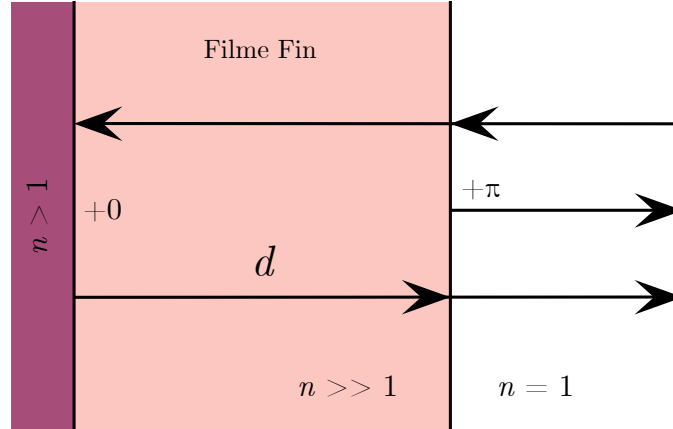


Figure 37. Thin Film.

We want to calculate the interference of a wave arriving at the thin film and the wave that bounces off the surface between the thin film and the second surface at $n > 1$ (but smaller than n_{film}). The simple path accumulates a phase of $\varphi_1 = \pi$, due to the index of refraction $n_{\text{film}} > n_{\text{air}} = 1$. The longer path accumulates a phase

$$\varphi_2 = 2\frac{n\omega}{c}d + 0 \quad (7.4)$$

Thus, for m integer, constructive interference requires

$$2\frac{n\omega}{c}d = (2m+1)\pi \Rightarrow d = \frac{2m+1}{2n\omega}c\pi = \lambda\frac{2m+1}{4n} \quad (7.5)$$

Destructive interference requires

$$2\frac{n\omega}{c}d = 2m\pi \Rightarrow d = \lambda\frac{m}{2n} \quad (7.6)$$

7.3 Two Rectangular Slits

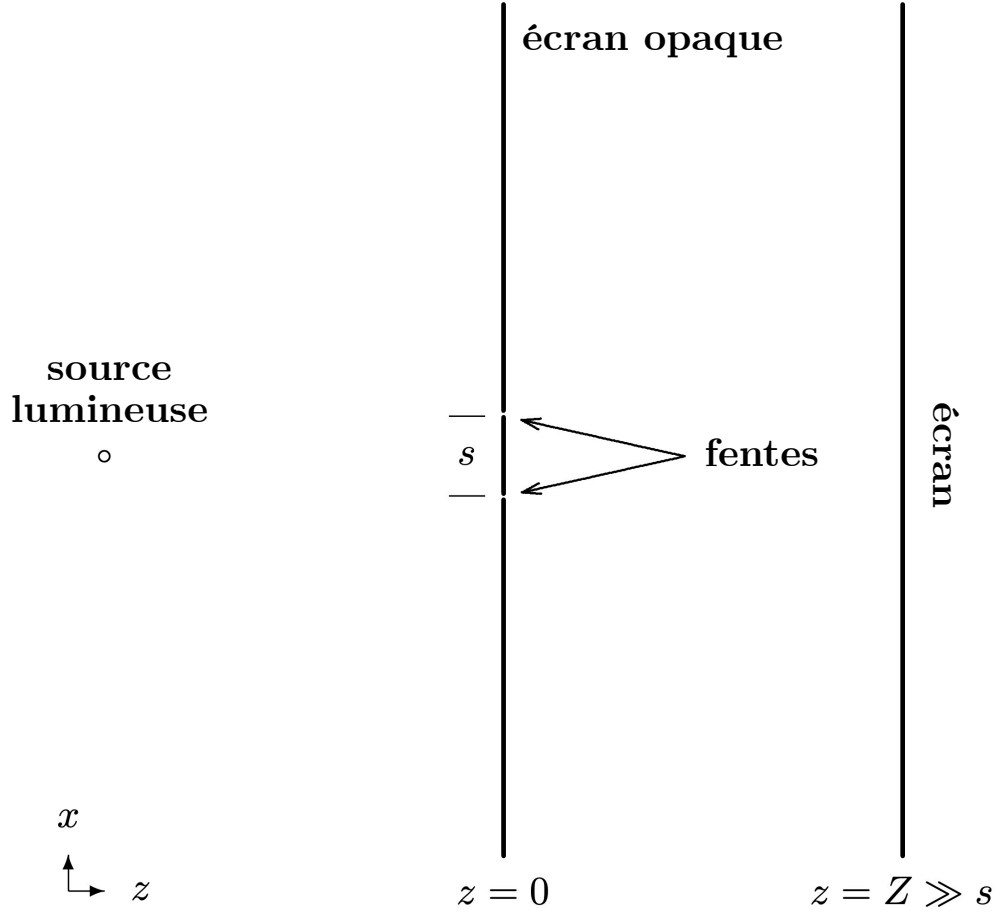


Figure 38. Two slits.

Consider two screens and a light source, as drawn above. If the slits are sufficiently small, we can compare the slits to point light sources, and thus we obtain the same antenna interference phenomenon. Namely, the interference will be governed by

$$\Delta\varphi = 2m\pi \Rightarrow \sin\theta = \frac{2m\pi}{s} \Rightarrow x_{\text{constr}} = \frac{2m\pi}{s}Z \quad \text{if } \sin\theta \simeq \tan\theta \quad (7.7)$$

Similarly, the points of destructive interference will be

$$\Delta\varphi = (2m+1)\pi \Rightarrow \sin\theta = \frac{(2m+1)\pi}{s} \Rightarrow x_{\text{destr}} = \frac{(2m+1)\pi}{s}Z \quad \text{if } \sin\theta \simeq \tan\theta \quad (7.8)$$

If the slit size (b) is appreciable, one must include a *Fraunhofer* term, or far-field diffraction. For a slit, this is obtained by considering the slit as a continuous distribution of point sources, such that

$$dE = \frac{A}{r_1} e^{i(kr_1 - \omega t)} dx \quad (7.9)$$

for $r_1 = r - x \sin \theta$ integrated from $-b/2 \rightarrow b/2$:

$$E \simeq \frac{A}{r} e^{i(kr - \omega t)} \int_{-b/2}^{b/2} e^{-ikx \sin \theta} dx = \frac{2A}{r} e^{i(kr - \omega t)} \frac{1}{k \sin \theta} \sin \left(\frac{kb}{2} \sin \theta \right) \quad (7.10)$$

Let $\psi' = kb \sin \theta$;

$$E = \frac{A}{r} e^{i(kr - \omega t)} b \operatorname{sinc}(\psi'/2\pi) \quad (7.11)$$

where

$$\operatorname{sinc}(x) \equiv \frac{\sin \pi x}{\pi x} \quad (7.12)$$

Thus, the intensity becomes

$$I_{\text{single}} \propto \operatorname{sinc}^2 \left(\frac{b \sin \theta}{\lambda} \right) \quad (7.13)$$

Thus, for two slits, this does not change, and thus

$$I \propto \cos^2 \left(\frac{\pi s \sin \theta}{\lambda} \right) \operatorname{sinc}^2 \left(\frac{b \sin \theta}{\lambda} \right) \quad (7.14)$$

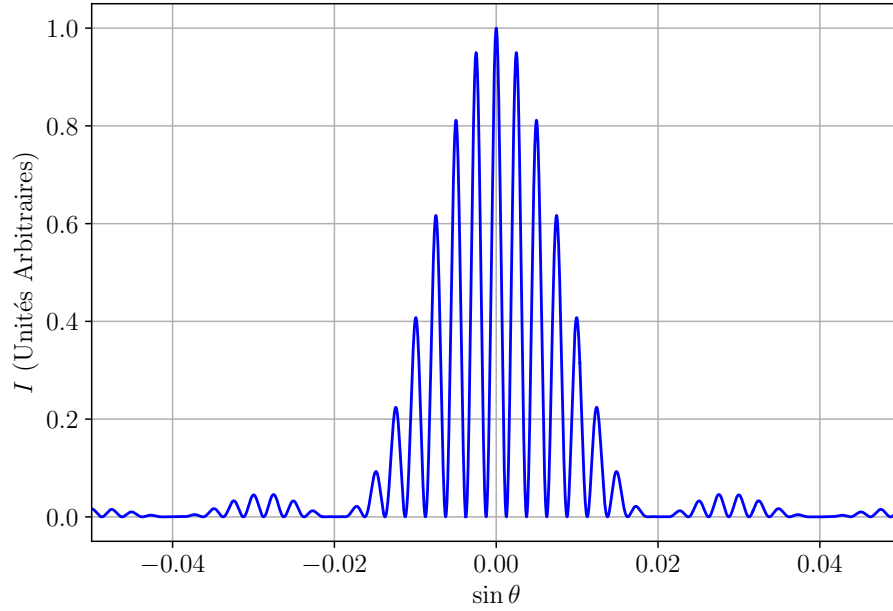


Figure 39. Two slits: $s = 5 \cdot 10^{-5}$, $b = 5 \cdot 10^{-6}$, for $\lambda = 10^{-7}$ (all units in m).

7.4 n Rectangular Slits

We can generalize to n slits. Let n slits have size b and spacing d . We can apply the same small angle approximation, such that the phase difference between light from two adjacent slits is

$$\Delta\varphi_{j,j-1} = kd \sin \theta = \frac{2\pi d \sin \theta}{\lambda} \quad (7.15)$$

Thus, we add the electric field resulting from the n slits, each with its phase:

$$E_{\text{net}} = \sum E_j = E_0 \sum \left(e^{2\pi i d \sin \theta / \lambda} \right)^j = E_0 \frac{1 - \exp \left(i \frac{2\pi n d \sin \theta}{\lambda} \right)}{1 - \exp \left(i \frac{2\pi d \sin \theta}{\lambda} \right)} \quad (7.16)$$

Thus, the intensity will be proportional to

$$I \propto \left| \frac{1 - \exp\left(i \frac{2\pi n d \sin \theta}{\lambda}\right)}{1 - \exp\left(i \frac{2\pi d \sin \theta}{\lambda}\right)} \right|^2 \quad (7.17)$$

We also add the term that describes the effect of slit size:

$$I \propto \text{sinc}^2\left(\frac{b \sin \theta}{\lambda}\right) \quad (7.18)$$

Thus, for n slits, for $\sin \theta \simeq \tan \theta$, the intensity on a screen a distance Z from the slits as a function of the vertical axis (parallel to that of the slits) is given by

$$I_n(X) = I_0 \left| \frac{1 - \exp\left(i \frac{2\pi n d X}{Z \lambda}\right)}{1 - \exp\left(i \frac{2\pi d X}{Z \lambda}\right)} \right|^2 \text{sinc}^2\left(\frac{b X}{Z \lambda}\right) \quad (7.19)$$

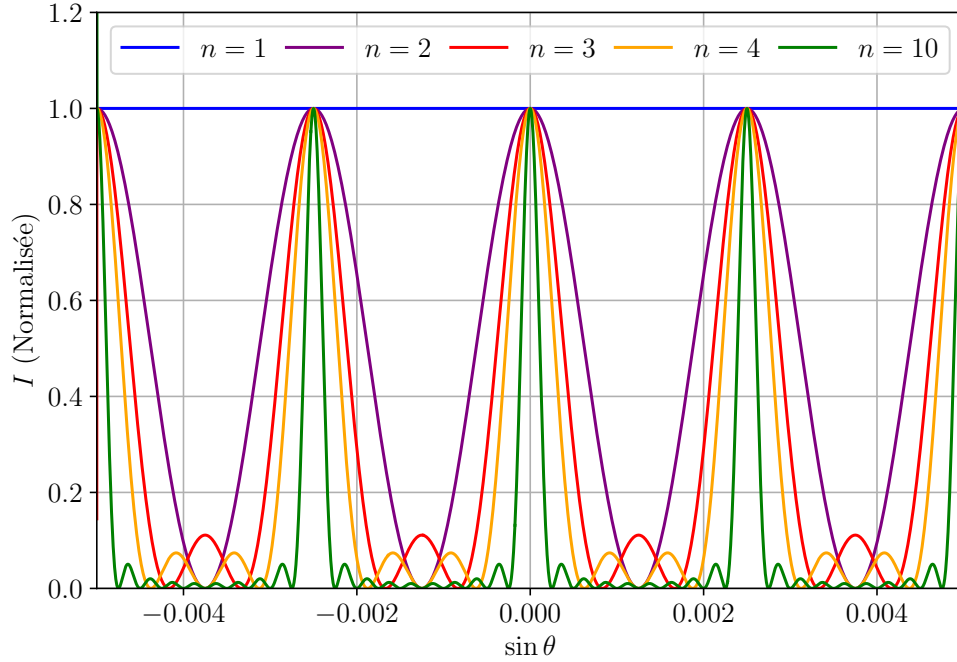


Figure 40. Interference from n slits.

Perhaps there is a general procedure for arbitrary slits.

7.5 Fourier Optics

Let a set of slits along the $\mathbf{e}_1$ axis be such that the function $f(x)$ corresponds to the transmittance of the slits: a slit of size b centered at the origin would be

$$f(x) = \text{rect}\left(\frac{x}{b}\right) \quad (7.20)$$

Let a screen be a distance Z from the slits. The distance between a point x on the slit plane and a point X on the screen plane is

$$\sqrt{(X-x)^2 + Z^2} \simeq \sqrt{X^2 + Y^2 + Z^2} - \frac{xX}{\sqrt{X^2 + Y^2 + Z^2}} \quad (7.21)$$

Let $R = \sqrt{(X-x)^2 + Z^2}$ and $\Delta l = -xX/R$. Thus, the accumulated phase is

$$e^{ik(R+\Delta l)} \quad (7.22)$$

If we integrate over all the slits (thus from $-\infty \rightarrow \infty$ along x), we have

$$E(X) = \int_{-\infty}^{\infty} f(x) e^{ik(R+\Delta l)} dx = \int_{-\infty}^{\infty} f(x) e^{-ixXk/R} dx = \mathcal{F}\{f\}|_{\omega=Xk/R} \quad (7.23)$$

In two dimensions, we have

$$E(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-i(xX+yY)k/R} dx dy = \mathcal{F}^2\{f\}|_{\omega_1=\frac{Xk}{R}, \omega_2=\frac{Yk}{R}} \quad (7.24)$$

This is the Fourier transform of the slits! We can thus apply this for more complicated slits.

7.5.1 n Slits via Fourier

Let there be two slits: we thus write

$$f(x) = \text{rect}\left(\frac{x}{b} + \frac{d}{2b}\right) + \text{rect}\left(\frac{x}{b} - \frac{d}{2b}\right) \quad (7.25)$$

Thus,

$$E(X) = \mathcal{F}\left\{\text{rect}\left(\frac{x}{b} + \frac{d}{2b}\right) + \text{rect}\left(\frac{x}{b} - \frac{d}{2b}\right)\right\}\Big|_{\omega=Xk/R} \quad (7.26)$$

$$\propto \text{sinc}\left(\frac{b\omega}{2\pi}\right) \left(e^{i\omega d/2} + e^{-i\omega d/2}\right)\Big|_{\omega=Xk/R} \quad (7.27)$$

$$= \text{sinc}\left(\frac{bXk}{2\pi R}\right) \left(e^{iXkd/2R} + e^{-iXkd/2R}\right) \quad (7.28)$$

We use $k = 2\pi/\lambda$:

$$E(X) \propto \text{sinc}\left(\frac{bX}{R\lambda}\right) \left(e^{iXd\pi/\lambda R} + e^{-iXd\pi/\lambda R}\right) \quad (7.29)$$

We write in terms of $\sin \theta = X/R$:

$$E(X) \propto \text{sinc}\left(\frac{b \sin \theta}{\lambda}\right) \left(e^{i\pi d \sin \theta / \lambda} + e^{-i\pi d \sin \theta / \lambda}\right) \propto \text{sinc}\left(\frac{b \sin \theta}{\lambda}\right) \cos\left(\frac{\pi d \sin \theta}{\lambda}\right) \quad (7.30)$$

Thus,

$$I(X) \propto \text{sinc}^2\left(\frac{b \sin \theta}{\lambda}\right) \cos^2\left(\frac{\pi d \sin \theta}{\lambda}\right) \quad (7.31)$$

This is exactly the result from before.

7.5.2 Circular Slits

Let a circular slit of radius ρ be such that

$$f(x, y) = \begin{cases} 1 & x^2 + y^2 \leq \rho^2 \\ 0 & x^2 + y^2 = \rho^2 \end{cases} \quad (7.32)$$

The diffraction pattern will thus be

$$E(X, Y) = \mathcal{F}^2\{f\}|_{\omega_1=\frac{Xk}{R}, \omega_2=\frac{Yk}{R}} \quad (7.33)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-i(xX+yY)k/R} dx dy \quad (7.34)$$

$$(7.35)$$

We write in polar coordinates:

$$E(R', \Theta) = \int_0^{2\pi} \int_0^\rho e^{-i(r \cos \theta R' \cos \Theta + r \sin \theta R' \sin \Theta)k/R} r dr d\theta \quad (7.36)$$

Unfortunately we can only evaluate this function numerically. In principle, this function would not depend on Θ due to the symmetry of the problem. We can draw the intensity:



Figure 41. Airy Pattern. The dashed blue line represents $r_0 = 1.22Z\lambda/2\rho$.

We can approximate the first zero diffraction radius:

$$r_0 = 1.22 \frac{Z\lambda}{2\rho} \quad (7.37)$$

This will be useful for angular resolution.

7.5.3 Rectangular Slits

We consider another case – a rectangular slit. Let

$$f(x, y) = \begin{cases} 1 & \langle x, y \rangle \in [-a_x, a_x] \times [-a_y, a_y] \\ 0 & \text{otherwise} \end{cases} \quad (7.38)$$

Thus,

$$E(X, Y) = \mathcal{F}^2\{f\}|_{\omega_1 = \frac{Xk}{R}, \omega_2 = \frac{Yk}{R}} \quad (7.39)$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x) e^{-i(xX+yY)k/R} dx dy \quad (7.40)$$

$$= \left(\int_{-a_x}^{a_x} e^{-ixXk/R} dx \right) \left(\int_{-a_y}^{a_y} e^{-iyYk/R} dy \right) \quad (7.41)$$

$$\propto \text{sinc}\left(\frac{2a_x X}{Z\lambda}\right) \text{sinc}\left(\frac{2a_y Y}{Z\lambda}\right) \quad (7.42)$$

The intensity is given by

$$I(X, Y) \propto \text{sinc}^2\left(\frac{2a_x X}{Z\lambda}\right) \text{sinc}^2\left(\frac{2a_y Y}{Z\lambda}\right) \quad (7.43)$$

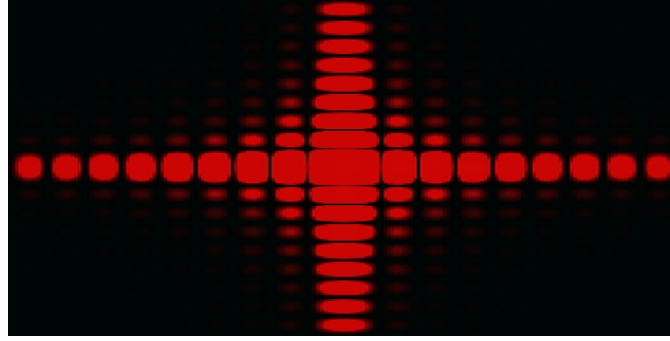


Figure 42. Rectangular diffraction.

7.6 Angular Resolution

It is possible to use Airy patterns to characterize the minimum angle between two distant light sources such that their images are distinct:

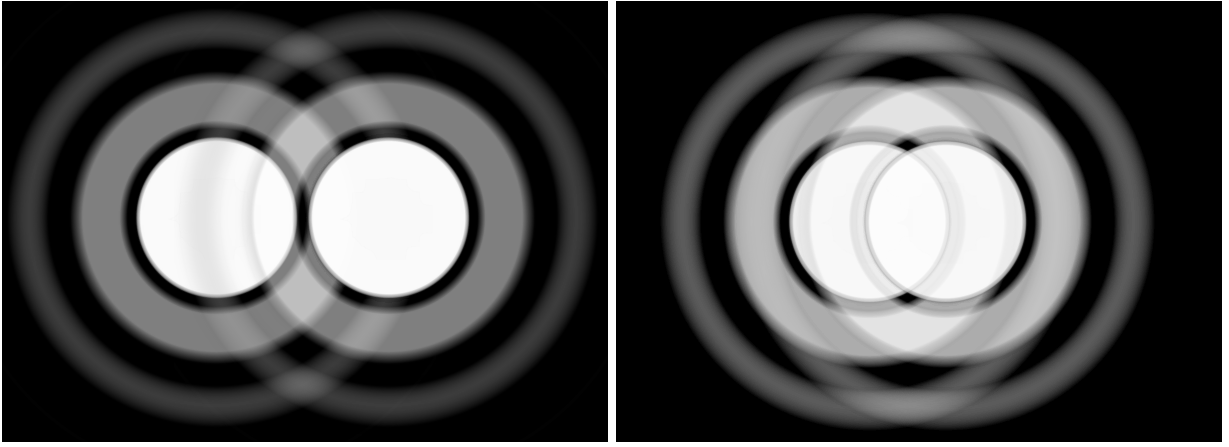


Figure 43. Critical distance between Airy patterns such that images are distinct.

We define the *angular resolution* of an optical system as the critical angle where the first minimum of one Airy pattern is aligned with the central maximum of the second.

Definition 7.6.1 (Rayleigh Criterion)

$$\Delta\theta = 1.22 \frac{\lambda}{D} \quad (7.44)$$

where D is the diameter of the aperture, λ is the wavelength, and $\Delta\theta$ is the angle between the two rays that contain the optical information.

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8.03 Document Captions

Figure 1:

"Équilibre" -> Equilibrium

Figure 6:

"unités de" -> units of

"Fréquence de" -> Frequency of

Figure 10:

"Masse" -> Mass

Figure 11:

"Position non-déformée" -> Non-deformed Position

"Position déformée" -> Deformed Position

Figure 13:

"Point d'Interêt" -> Point of Interest

Figures 14-17:

"LIBRE" -> FREE

"FIXE" -> FIXED

Figure 30:

"Région *" -> Region *

"où la particule est maintenant au repos" -> where the particle is now at rest

"où la particule aurait été si elle ne s'était pas arrêtée" -> where the particle would have been should it have continued

Figure 36:

"miroir" -> mirror

"Régime d'Interférence" -> Interference Regime/Region

Figure 37:

"Filme Fin" -> Thin Film

Figure 38:

"source lumineuse" -> Luminous Source (i.e. point light source)

"écran opaque" -> opaque screen/wall

"fentes" -> slits

"écran" -> screen

Figure 39:

"Unités Arbitraires" -> Arbitrary Units

Figure 40:

"Normalisée" -> Normalized